

EMISSION FACTORS

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This annex summarizes the development and selection of emission factors (EF) used to estimate Canada's annual greenhouse gas (GHG) inventory. Details on sector-specific methodological use of these factors are presented in [Annex 3](#).

A6.1. Fuel Combustion

A6.1.1. Natural Gas and Natural Gas Liquids

A6.1.1.1. Carbon Dioxide (CO₂)

CO₂ emission factors for fossil fuel combustion depend primarily on fuel properties such as carbon content, density and heating value and, to a lesser extent, on the combustion technology.

For natural gas, there are two principal types combusted in Canada: marketable fuel (processed for commercial sale) and non-marketable fuel (unprocessed, for internal use). There are regional variations in marketable and non-marketable natural gas use, with nine regions consuming marketable fuel and seven regions consuming non-marketable fuel.

For the period 1990 to 1998, provincial and territorial emission factors for marketable natural gas ([Table A6.1–1](#)) were developed using data from the chemical analysis of representative natural gas samples collected in 1998 (McCann, 2000). Both imported and domestic natural gas were included, where applicable, in the mix of gas samples used for chemical analysis.

For the period 2005 to 2018, an industry expert was contracted to work with the natural gas transmission and distribution industry to define provincial natural gas flow networks and to collect historical data. Over 500 000 data points were collected for these years at key points, such as transmission lines, interprovincial transfer locations, storage locations, import and export sites, and consumer distribution networks, where volumetric flow and composition data were measured and available. Using this data, the emission factors and energy content values of marketable natural gas were updated for the years 2005 to 2023 ([Table A6.1–1](#)) to reflect a gradual increase in complex hydrocarbons in the natural gas stream that offset the approximately 2% decrease in methane over the same time period. Data collected by the contractor underwent further analysis (ECCC, 2021a) and this showed that since it was last measured in 1998 the percent methane composition in the natural gas stream had changed in the period from 1999 to 2004. This resulted in a step change in natural gas parameters and emissions from 2004 to 2005. To minimize the effect of this step change, ECCC revised the natural gas parameters for these intermediate years, 1999 to 2004, using a simple linear interpolation.

Non-marketable natural gas emission factors for Alberta ([Table A6.1–2](#)) were updated using over 400 000 raw gas samples, ranging in date from 1913 to 2016, obtained from the Alberta Energy Regulator (AER) by the Energy and Emissions Research Laboratory (EERL) of Carleton University (EERL, 2020). EERL compiled and analysed the data to provide average gas compositions by township. Gas composition data by township were weighted by township-level fuel gas consumption volumes obtained from Petrinex (2010–2023), to produce annual average CO₂ emission factors for the province.

Non-marketable natural gas is produced and consumed at oil and gas facilities offshore of Newfoundland and Labrador. Non-marketable natural gas CO₂ emission factors for these offshore facilities were derived using emissions data reported to the GHG Reporting Program (ECCC, 2024a) and fuel and flare volumes reported by facilities to the Canada-Newfoundland and Labrador Offshore Petroleum Board (CNLOPB, 2024).

Generally, non-marketable natural gas emission factors (Table A6.1–2) are higher than those of marketable natural gas (Table A6.1–1) because of their raw/unprocessed nature where complex hydrocarbons are not removed prior to usage. In addition to methane, the composition of marketable and non-marketable natural gas includes ethane, propane, butane, and other more complex hydrocarbons.

CO₂ emission factors (Table A6.1–4) for natural gas liquids (NGL), such as ethane, propane and butane, were developed based on chemical analysis data for marketable fuels (McCann, 2000).

A6.1.1.2. Methane (CH₄)

Emissions of CH₄ from fuel combustion are technology-dependent. Sectoral emission factors (Table A6.1–3, Table A6.1–4, and Table A6.1–5) have been developed based on technologies typically used in Canada. For most sources, the factors were developed based on a broad review of emission factors for combustion technologies (SGA Energy, 2000). The generic emission factor for producer consumption of natural gas in Table A6.1–3 was developed based on a technology split for the upstream oil and gas industry (CAPP, 1999) and technology-specific emission factors from the U.S. EPA report AP 42 (U.S. EPA, 1996). Province-specific values in Table A6.1–4 were calculated using the same approach, but the fuel volumes used to weight the emissions factors were updated to reflect the annual distribution of producer consumption within the upstream oil and gas industry.

A6.1.1.3. Nitrous Oxide (N₂O)

Emissions of N₂O from fuel combustion are technology-dependent. Emission factors (Table A6.1–3 and Table A6.1–5) have been developed based on technologies typically used in Canada. The factors were developed from an analysis of combustion technologies and a review of their emission factors (SGA Energy, 2000).

Table A6.1–1 CO₂ Emission Factors for Marketable Natural Gas

Year	Emission Factors (g/m ³) ^a													
	Canada	BC	AB	SK	MB	QC	ON	NB	NS	PE	NL	YT	NT ^b	NU
1990	1 899	1 926	1 928	1 829	1 886	1 887	1 888	1 901	1 901	1 901	1 901	1 901	2 466	NO
1995	1 899	1 926	1 928	1 829	1 886	1 887	1 888	1 901	1 901	1 901	1 901	1 901	2 466	NO
2000	1 895	1 918	1 923	1 839	1 880	1 880	1 881	1 884	1 884	1 884	1 884	1 900	2 466	1 900
2005	1 883	1 898	1 911	1 864	1 864	1 864	1 865	1 864	1 864	1 864	1 867	1 898	2 466	1 898
2006	1 878	1 896	1 917	1 855	1 855	1 854	1 855	1 855	1 855	1 855	1 862	1 896	2 466	1 896
2007	1 881	1 901	1 910	1 857	1 857	1 858	1 859	1 858	1 858	1 858	1 856	1 901	2 466	1 901
2008	1 887	1 906	1 910	1 863	1 863	1 865	1 865	1 864	1 864	1 864	1 860	1 906	2 466	1 906
2009	1 889	1 900	1 920	1 861	1 861	1 865	1 863	1 862	1 862	1 862	1 865	1 900	2 466	1 900
2010	1 891	1 902	1 923	1 860	1 862	1 868	1 869	1 865	1 865	1 865	1 864	1 902	2 466	1 902
2011	1 893	1 900	1 921	1 866	1 866	1 870	1 871	1 869	1 869	1 869	1 869	1 900	2 466	1 900
2012	1 905	1 899	1 931	1 874	1 873	1 882	1 888	1 880	1 880	1 880	1 873	1 899	1 899	1 899
2013	1 910	1 909	1 933	1 875	1 873	1 884	1 896	1 887	1 887	1 887	1 886	1 909	1 909	1 909
2014	1 912	1 912	1 930	1 878	1 875	1 882	1 896	1 887	1 887	1 887	1 892	1 912	1 912	1 912
2015	1 924	1 939	1 943	1 881	1 905	1 911	1 911	1 912	1 912	1 912	1 900	1 939	1 939	1 939
2016	1 932	1 941	1 954	1 897	1 896	1 907	1 915	1 908	1 908	1 908	1 914	1 941	1 941	1 941
2017	1 936	1 955	1 955	1 900	1 903	1 915	1 916	1 910	1 910	1 910	1 914	1 955	1 955	1 955
2018	1 944	1 966	1 962	1 920	1 915	1 926	1 921	1 919	1 919	1 919	1 918	1 966	1 966	1 966
2019	1 944	1 966	1 962	1 920	1 915	1 926	1 921	1 919	1 919	1 919	1 921	1 966	1 966	1 966
2020	1 944	1 966	1 962	1 920	1 915	1 926	1 921	1 919	1 919	1 919	1 921	1 966	1 966	1 966
2021	1 944	1 966	1 962	1 920	1 915	1 926	1 921	1 919	1 919	1 919	1 921	1 966	1 966	1 966
2022	1 944	1 966	1 962	1 920	1 915	1 926	1 921	1 919	1 919	1 919	1 921	1 966	1 966	1 966
2023	1 944	1 966	1 962	1 920	1 915	1 926	1 921	1 919	1 919	1 919	1 919	1 966	1 966	1 966

Notes:

The term "marketable" applies to fuel consumed by the Utility, Industry, Residential, Commercial, and Transport subsectors.

NO = Not occurring

a. Years 1990–1998, McCann (2000). Years 1999–2023, ECCC (2021a).

b. Prior to 2012, natural gas consumption was regional non-marketable natural gas. Since 2012, marketable natural gas has been imported from BC.

Table A6.1–2 **CO₂ Emission Factors for Non-Marketable Natural Gas**

Year	Emission Factors (g/m ³)												
	BC	AB ^a	SK	MB	QC	ON	NB	NS	PE	NL ^b	YT	NT	NU
1990	2 162	2 080	2 441	2 401	NO	2 401	2 401	2 494	NO	NO	2 401	2 466	NO
1995	2 162	2 080	2 441	2 401	NO	2 401	2 401	2 494	NO	NO	2 401	2 466	NO
2000	2 162	2 080	2 441	2 401	NO	2 401	2 401	2 494	NO	x	2 401	2 466	NO
2005	2 162	2 080	2 441	2 401	NO	2 401	2 401	2 494	NO	2 299	2 401	2 466	NO
2006	2 162	2 080	2 441	2 401	NO	2 401	2 401	2 494	NO	2 298	2 401	2 466	NO
2007	2 162	2 080	2 441	2 401	NO	2 401	2 401	2 494	NO	2 294	2 401	2 466	NO
2008	2 162	2 080	2 441	2 401	NO	2 401	2 401	2 494	NO	2 327	2 401	2 466	NO
2009	2 162	2 080	2 441	2 401	NO	2 401	2 401	2 494	NO	2 309	2 401	2 466	NO
2010	2 162	2 054	2 441	2 401	NO	2 401	2 401	2 494	NO	2 276	2 401	2 466	NO
2011	2 162	2 062	2 441	2 401	NO	2 401	2 401	2 494	NO	2 220	2 401	2 466	NO
2012	2 162	2 068	2 441	2 401	NO	2 401	2 401	2 494	NO	2 281	2 401	2 466	NO
2013	2 162	2 071	2 441	2 401	NO	2 401	2 401	2 494	NO	2 272	2 401	2 466	NO
2014	2 162	2 075	2 441	2 401	NO	2 401	2 401	2 494	NO	2 263	2 401	2 466	NO
2015	2 162	2 081	2 441	2 401	NO	2 401	2 401	2 494	NO	2 260	2 401	2 466	NO
2016	2 162	2 089	2 441	2 401	NO	2 401	2 401	2 494	NO	2 261	2 401	2 466	NO
2017	2 162	2 093	2 441	2 401	NO	2 401	2 401	2 494	NO	2 250	2 401	2 466	NO
2018	2 162	2 101	2 441	2 401	NO	2 401	2 401	2 494	NO	2 191	2 401	2 466	NO
2019	2 162	2 106	2 441	2 401	NO	2 401	2 401	2 494	NO	2 218	2 401	2 466	NO
2020	2 162	2 109	2 441	2 401	NO	2 401	2 401	2 494	NO	2 274	2 401	2 466	NO
2021	2 162	2 109	2 441	2 401	NO	2 401	2 401	2 494	NO	2 210	2 401	2 466	NO
2022	2 162	2 112	2 441	2 401	NO	2 401	2 401	2 494	NO	2 340	2 401	2 466	NO
2023	2 162	2 112	2 441	2 401	NO	2 401	2 401	2 494	NO	2 260	2 401	2 466	NO

Notes:

The term "non-marketable" applies to raw/unprocessed gas consumption, mainly by natural gas producers.

NO = Not occurring

x = confidential

a. Adapted from EERL (2020) using Petrinex (2010–) annual volumetric data

b. Derived using ECCC (2024a) and CNLOPB (2024)

All other provinces/territories from McCann (2000)

Table A6.1–3 **CH₄ and N₂O Emission Factors for Natural Gas**

Source	Emission Factor (g/m ³) ^a	
	CH ₄	N ₂ O
Electric Utilities	0.490	0.049
Industrial	0.037	0.033
Producer Consumption (Non-marketable)	6.4 ^b	0.060
Producer Consumption (Non-marketable) – Newfoundland and Labrador	0.490	0.060
Pipelines	1.9	0.050
Cement	0.037	0.034
Manufacturing Industries	0.037	0.033
Residential, Construction, Commercial/Institutional, Agriculture	0.037	0.035

Notes:

a. SGA Energy (2000).

b. Adapted from U.S. EPA (1996) and CAPP (1999). Not applicable in British Columbia, Alberta, and Saskatchewan.

Table A6.1–4 **CH₄ Emission Factors for Natural Gas Producer Consumption in Western Provinces**

Year	Emission Factors (g/m ³)						
	BC ^a	AB ^b	SK ^c	Year	BC ^a	AB ^b	SK ^c
1990	8.85	6.40	6.02	2007	9.15	8.52	4.67
1991	8.85	6.40	6.42	2008	9.04	8.55	4.55
1992	8.85	6.40	6.29	2009	9.03	8.51	4.44
1993	8.85	6.40	5.99	2010	8.91	8.52	4.10
1994	8.85	6.40	5.71	2011	8.80	8.58	3.80
1995	8.85	6.40	5.54	2012	8.66	8.32	3.60
1996	8.85	6.62	5.30	2013	8.59	8.33	3.44
1997	8.85	6.85	5.18	2014	8.61	8.32	4.04
1998	8.85	7.07	5.29	2015	8.41	8.22	3.81
1999	8.85	7.29	5.19	2016	8.47	8.32	3.89
2000	8.85	7.52	5.04	2017	8.39	8.39	3.89
2001	8.85	7.74	5.26	2018	8.41	8.25	4.16
2002	8.87	7.96	5.44	2019	8.44	8.16	4.55
2003	8.79	8.19	5.78	2020	8.59	8.23	4.82
2004	9.15	8.41	4.90	2021	8.73	8.07	4.49
2005	9.15	8.51	4.66	2022	8.83	8.21	4.09
2006	9.15	8.50	4.77	2023	9.20	8.22	4.20

Notes:

All EFs adapted from U.S. EPA (1996) and CAPP (1999) and weighted using annual fuel gas volumes taken from the following sources and grouped to align with CAPP (1999).

a. Years 2019–2023, BCER (2024). Years 2001–2018, BC (2019). Years 1990–2000, constant.

b. Years 2010–2023, Petrinex (2010–). Years 2004–2009, AER (2003–). Years 1996–2003, linear interpolation. Years 1990–1995, CAPP (1999).

c. Years 2013–2023, Petrinex (2013–). Years 1990–2012, SK MER (2012–).

Table A6.1–5 **Emission Factors for Natural Gas Liquids**

Source	Emission Factor (g/L)		
	CO ₂	CH ₄	N ₂ O
Propane			
Residential	1 515 ^a	0.027 ^b	0.108 ^b
All Other Uses	1 515 ^a	0.024 ^b	0.108 ^b
Ethane	986 ^a	0.024 ^b	0.108 ^b
Butane	1 747 ^a	0.024 ^b	0.108 ^b

Notes:

a. McCann (2000)

b. SGA Energy (2000)

A6.1.2. Refined Petroleum Products

A6.1.2.1. CO₂

CO₂ emission factors for fossil fuel combustion are dependent primarily on fuel properties and, to a lesser extent, on the combustion technology.

Emission factors (Table A6.1–6) have been developed for each major class of refined petroleum products (RPP) based on their heating value, carbon content and density (McCann, 2000), to ensure consistency with the 2006 *Intergovernmental Panel on Climate Change (IPCC) Guidelines for National Greenhouse Gas Inventories* (IPCC, 2006).

The composition of petroleum coke is process-specific. Factors have been developed for both refinery (catalytic cracker) derived cokes and coke used in upgrading facilities. These factors (Table A6.1–7) have been developed using emissions and energy content data provided by industry to the Canadian Energy and Emissions Data Centre (CEEDC).¹ The bulk of the coke consumed by refineries is catalytic cracker-derived, and the emission factor is a weighted average of petroleum

1 Griffin, B. 2024. Personal communication (email from Griffin B., CEEDC to Kay J., Physical Scientist, PIRD dated October 10, 2024). Canadian Energy and Emissions Data Centre (CEEDC).

coke and catalytic cracker coke emission factors. In the electricity generation sector in New Brunswick, petroleum coke is imported and mixed with coal prior to combustion. Carbon content data specific to this imported petroleum coke was used to develop a sector-specific petroleum emission factor, used only in New Brunswick. Since the data is facility-specific, the petroleum coke emission factor provided in [Table A6.1–6](#) is a weighted emission factor for Canada.

Emission factors for still gas ([Table A6.1–7](#)) from refining operations and upgrading facilities were also derived from data provided by industry and reported by CEEDC.

A6.1.2.2. CH₄

Emissions of CH₄ from fuel combustion are technology-dependent. Emission factors were developed ([Table A6.1–6](#)) based on technologies typically used in Canada. The factors were developed from an analysis of combustion technologies and a review of their emission factors (SGA Energy, 2000).

The emission factor for petroleum coke was assumed to be the same for both catalytic cracker-derived cokes and coke used in upgrading facilities.

The emission factor for still gas from upgraders ([Table A6.1–6](#)) was based on the 2006 IPCC default emission factor and was adapted using energy conversion factors provided by CEEDC. The still gas emission factors for refineries and other industries ([Table A6.1–9](#)) were based on the 2006 IPCC default emission factor, which was calculated on an annual basis using energy conversion factors provided by CEEDC.

A6.1.2.3. N₂O

Emissions of N₂O from fuel combustion are technology-dependent. Emission factors for RPP, except for petroleum coke, have been developed ([Table A6.1–6](#)) based on technologies typically used in Canada. The factors were developed from an analysis of combustion technologies and a review of their emission factors (SGA Energy, 2000).

Emission factors for petroleum coke ([Table A6.1–8](#)) were based on 2006 IPCC default emission factors and were calculated on an annual basis using energy conversion factors provided by Statistics Canada (2014) and CEEDC.

Table A6.1–6 Emission Factors for Refined Petroleum Products			
Source	Emission Factor (g/L)		
	CO ₂ ^a	CH ₄ ^b	N ₂ O ^c
Light Fuel Oil			
Electric Utilities	2 753	0.18	0.031
Industrial	2 753	0.006	0.031
Producer Consumption	2 670	0.006	0.031
Residential	2 753	0.026	0.006
Forestry, Construction, Public Administration and Commercial/Institutional	2 753	0.026	0.031
Heavy Fuel Oil			
Electric Utilities	3 156	0.034	0.064
Industrial	3 156	0.12	0.064
Producer Consumption	3 190	0.12	0.064
Residential, Forestry, Construction, Public Administration and Commercial/Institutional	3 156	0.057	0.064
Kerosene			
Electric Utilities	2 560 ^c	0.006	0.031
Industrial	2 560 ^c	0.006	0.031
Producer Consumption	2 560 ^c	0.006	0.031
Residential	2 560 ^c	0.026	0.006
Forestry, Construction, Public Administration and Commercial/Institutional	2 560 ^c	0.026	0.031
Diesel – Refineries and Others^d	2 681	0.078	0.022
Diesel – Upgraders^d	2 681	0.078	0.022
Petroleum Coke	See Table A6.1–7	0.12	See Table A6.1–8
Still Gas – Refineries and Others	See Table A6.1–7	See Table A6.1–9	0.00002
Still Gas – Upgraders	See Table A6.1–7	0.000039	0.00002
Motor Gasoline^e	2 307	0.100	0.02
Notes:			
a. McCann (2000); except Kerosene, Diesel and Motor Gasoline			
b. SGA Energy (2000); except Diesel and Motor Gasoline			
c. Assumed McCann (2000) aviation turbo-fuel emission factor			
d. CO ₂ from ECCC (2017b); CH ₄ and N ₂ O from Oak Leaf Environmental Inc. (2017)			
e. CO ₂ from ECCC (2017b); CH ₄ and N ₂ O adapted from IPCC (2006)			

Table A6.1–7 **CO₂ Emission Factors for Petroleum Coke and Still Gas**

Year	Emission Factor			
	Petroleum Coke (g/L)		Still Gas (g/m ³)	
	Upgrading Facilities ^a	Refineries and Others ^b	Upgrading Facilities ^a	Refineries and Others ^c
1990	2 965	3 140	2 310	1 740
1995	2 961	3 160	2 090	1 800
2000	2 903	3 090	2 120	1 680
2005	2 837	3 141	2 140	1 707
2006	2 813	3 150	2 140	1 741
2007	2 800	3 174	2 140	1 749
2008	2 813	3 193	2 140	1 690
2009	2 792	3 199	2 140	1 711
2010	2 817	3 261	2 140	1 825
2011	2 820	3 234	2 140	1 818
2012	2 802	3 240	2 140	1 718
2013	2 765	3 220	2 140	1 738
2014	2 794	3 227	2 140	1 741
2015	2 795	3 232	2 140	1 762
2016	2 781	3 236	2 140	1 781
2017	2 800	3 253	2 140	1 796
2018	2 769	3 235	2 140	1 847
2019	2 771	3 258	2 140	1 849
2020	2 769	3 246	2 140	1 770
2021	2 744	3 250	2 140	1 764
2022	2 729	3 199	2 140	1 773
2023	2 717	3 254	2 140	1 773

Notes:

a. CEEDC (2003)

b. Weighted emission factor based on Griffin B. 2024. Personal communication (email from Griffin B to Kay J, Physical Scientist, Pollutant Inventories and Reporting Division, dated October 10, 2024). Canadian Emissions and Energy Data Centre, and Samms J. 2020 Personal Communication (email from Samms J to Tracey K, Senior Program Engineer, Pollutant Inventories and Reporting Division, dated December 8, 2020).

c. Griffin B. 2024. Personal communication (email from Griffin B to Kay J, Physical Scientist, Pollutant Inventories and Reporting Division, dated October 10, 2024). Canadian Emissions and Energy Data Centre.

Table A6.1–8 **N₂O Emission Factors for Petroleum Coke**

Year	Emission Factor	
	Petroleum Coke (g/m ³)	
	Upgrading Facilities ^{a, b}	Refineries and Others ^{a, c}
1990	18.2	21.9
1995	18.9	22.9
2000	20.1	23.5
2005	19.5	23.0
2006	19.4	23.1
2007	19.3	23.1
2008	19.4	23.1
2009	19.2	23.2
2010	19.4	23.2
2011	19.4	23.2
2012	19.3	23.4
2013	19.0	23.4
2014	19.2	23.4
2015	19.2	23.6
2016	19.1	23.5
2017	19.3	23.6
2018	19.1	23.5
2019	19.1	23.6
2020	19.1	23.6
2021	18.9	23.6
2022	18.8	23.3
2023	18.7	23.6

Notes:

a. Adapted from IPCC (2006)

b. Energy content from Statistics Canada (2014)

c. Energy content from Griffin B. 2024. Personal communication (email from Griffin B to Kay J, Physical Scientist, Pollutant Inventories and Reporting Division, dated October 10, 2024). Canadian Emissions and Energy Data Centre.

Table A6.1–9 CH₄ Emission Factors for Still Gas (Refineries and Others)

Year	Emission Factor ^a
	g/m ³
1990–1996	0.033
1997	0.032
1998	0.031
1999	0.033
2000	0.034
2001	0.034
2002	0.033
2003	0.033
2004	0.033
2005	0.032
2006	0.032
2007	0.032
2008	0.032
2009	0.032
2010	0.032
2011	0.033
2012	0.031
2013	0.031
2014	0.031
2015	0.033
2016	0.032
2017	0.033
2018–2023	0.032

Note:

a. Adapted from IPCC (2006) using energy content taken from Griffin B. 2019. Personal communication (email from Griffin B to Tracey K, Senior Program Engineer, Pollutant Inventories and Reporting Division, dated Sept 26, 2019). Canadian Emissions and Energy Data Centre.

A6.1.3. Coal and Coal Products

A6.1.3.1. CO₂

CO₂ emission factors for coal combustion depend largely on the properties of the fuel and, to a lesser extent, on the combustion technology. Coal emission factors (Table A6.1–10) were developed for each province based on the rank of the coal and the region of supply. Emission factors were based on data from chemical analysis of coal samples for electric utilities, which account for most of the coal consumption.

ECCC has been developing coal emission factors over time, starting in 2012 with the analysis of over 3000 analytical samples for a variety of coal types and producing/consuming regions. The analysis and uncertainty calculations were conducted using the @Risk software package. The coal emission factors are presented with uncertainty estimates since the supply and quality of coal can vary over time. The average coal carbon and moisture content for each coal type were used to develop CO₂ emission factors.

In 2015, an additional study to determine country-specific coal oxidation factors and further investigate the carbon content of coal burned at electric generation facilities was conducted for ECCC by GHD Limited. In 2021, ECCC collected further carbon content coal data from electric generation facilities that did not participate in the 2015 study. Based on an analysis of the data collected since 2012, ECCC has developed rank and region-specific emission and oxidation factors as well as uncertainty estimates for all but anthracite (ECCC, 2023a).

Factors for anthracite imported from the United States are from a study done for Environment Canada (EC) on CO₂ emission factors, based on data collected on coal consumption in Canada in 1998 (McCann, 2000). All coal emission factors in Table A6.1–10 now incorporate Canada-specific oxidation factors (ECCC, 2023a).

Table A6.1–10 **CO₂ Emission Factors for Coal**

Province	Coal Type	Source	Emission Factor (kg CO ₂ / tonne) ^{a, b}			Oxidation Factor ^b (%)	Moisture (wt %)
			Mean	Uncertainty (95% CI) ^c			
				Low	High		
NL, NS, PE, QC (Prior to 2000)	Canadian Bituminous	NS	2 344	-33%	22%	98.78	3.2
NL, NS, PE, QC (2000 onward)	Canadian Bituminous	AB	2 212	-26%	26%	98.78	7.7
NB (Prior to 2010)	Canadian Bituminous	NB	2 333	-14%	14%	98.78	3.2
NB (2010 onward)	Canadian Bituminous	AB	2 212	-26%	26%	98.78	7.7
AB, BC, MB, ON, SK	Canadian Bituminous	AB	2 212	-26%	26%	98.78	7.7
NB, NS, PE and NL	Foreign Bituminous	Non-U.S.	2 571	-7%	7%	98.78	8.3
MB	Foreign Bituminous	U.S. (Pennsylvania)	2 651	-7%	7%	98.78	N/A
ON	Foreign Bituminous	U.S. (Pennsylvania)	2 651	-7%	7%	98.38	N/A
QC, AB, BC	Foreign Bituminous	U.S. (Pennsylvania)	2 662	-7%	7%	98.78	N/A
All Provinces and Territories, except SK	Lignite	SK	1 463	-13%	13%	99.92	24
SK	Lignite	SK	1 463	-13%	13%	99.61	36
QC, ON, MB	Sub-bituminous	Foreign	1 865	-8%	8%	99.84	24
NS, PE	Sub-bituminous	Foreign	1 743	-8%	8%	99.84	24
AB, SK, BC	Sub-bituminous	AB	1 775	-11%	11%	99.36	21
NB (2010, 2011, 2013, 2014)	Sub-bituminous	Foreign	2 189	-2%	2%	99.84	19
NB (2012, 2015 onward)	Sub-bituminous	Foreign	2 352	-2%	2%	99.84	12
All Provinces and Territories	Anthracite	--	3 097	-6%	6%	98.78	N/A

Notes:

CI = confidence interval

N/A = Not available

a. Factors presented on a "wet basis." Moisture content shown is that for the "weighted average" emission factor.

b. ECCC (2023)

c. 95% Confidence Intervals, which were determined through statistical analysis of Canadian coal data.

Table A6.1–11 **CO₂ Emission Factors for Coke Oven Gas and Coke**

Coal Product – Fuel Type	Emission Factor
Coke Oven Gas ^a	687 g/m ³
Coke ^b	3 173 g/kg

Notes:

a. McCann (2000)

b. CRA (2014)

Coke and coke oven gas emission factors are presented in [Table A6.1–11](#). The coke emission factor was developed from an iron and steel industry study completed in 2014 (CRA, 2014). It is representative of coke use in the cement, non-ferrous metal and other manufacturing industries. The coke oven gas emission-factor value is from McCann (2000) and represents use in the iron and steel industry.

A6.1.3.2. **CH₄**

Emissions of CH₄ from fuel combustion are technology-dependent. Emission factors for sectors ([Table A6.1–12](#)) have been developed based on technologies typically used in Canada. The factors were developed from an analysis of combustion technologies and a review of their emission factors (SGA Energy, 2000).

A6.1.3.3. **N₂O**

Emissions of N₂O from fuel combustion are technology-dependent. Emission factors for sectors ([Table A6.1–12](#)) have been developed based on technologies typically used in Canada. The emission factors were developed from an analysis of combustion technologies and a review of their emission factors (SGA Energy, 2000).

Table A6.1–12 **CH₄ and N₂O Emission Factors for Coal, Coke and Coke Oven Gas**

Source	Emission Factor	
	CH ₄	N ₂ O
	g/kg	
Coal		
Electric Utilities	0.02	0.03
Industry and Heat & Steam Plants	0.03	0.02
Residential, Public Administration	4.00	0.02
Coke	0.03	0.02
	g/m ³	
Coke Oven Gas	0.04	0.04
Note: Source: SGA Energy (2000)		

A6.1.4. Fugitive Emission Factors for Coal Mining

The factors in Table A6.1–13 are for fugitive emissions from coal mining only. Although derived from measurements at individual mines or coal seams, these emission factors are aggregated and weighted, province-wide averages for a given mine type. These weighted emission factors are updated yearly to account for the change in production of coal types, at individual mines. They should only be applied to total provincial gross (not net) production quantities of coal mined which includes small quantities of minerals, stone and other inert materials mined with the coal, but later removed before sale or consumption.

Table A6.1–13 **Provincially Weighted Coal Mining Fugitive Emission Factors (t CH₄/kt coal mined)**

	Surface Coal Mines				
	Bituminous			Sub-bituminous	Lignite
	AB	BC	NS	AB	SK
1990	0.58	0.82	C	0.18	0.07
1995	0.58	0.82	0.07	0.21	0.07
2000	0.59	0.82	0.07	0.21	0.07
2005	0.51	0.87	0.07	0.21	0.07
2006	0.48	0.86	0.07	0.21	0.07
2007	0.47	0.87	C	0.21	0.07
2008	0.47	0.85	C	0.21	0.07
2009	0.43	0.85	C	0.20	0.07
2010	0.44	0.86	C	0.20	0.07
2011	0.42	0.84	C	0.21	0.07
2012	0.45	0.85	C	0.21	0.07
2013	0.46	0.84	C	0.22	0.07
2014	0.52	0.84	C	0.20	0.07
2015	0.55	0.86	C	0.20	0.07
2016	C	0.87	C	0.21	0.07
2017	C	0.86	C	0.22	0.07
2018	0.51	0.92	C	0.23	0.07
2019	0.47	0.93	C	0.24	0.07
2020	0.36	0.93	C	0.21	0.07
2021	C	0.93	C	0.15	0.07
2022	0.31	0.92	C	0.15	0.07
2023	0.31	0.92	C	0.15	0.07
Note: C = Confidential Adapted from King (1994), and Cheminfo and Clearstone Engineering Ltd (2014). Weighted emission factors for underground coal mines in Nova Scotia, Alberta, and British Columbia are either confidential or there are no mines operating due to closures.					

A6.1.5. Other Fuels

A6.1.5.1. CO₂

Alternative fuels such as tires, shingles, plastics, refuse, and waste oil, solvents and other fossil-based wastes, along with dried sewage sludge, paper, animal meal, animal fat and other biogenic wastes are used in the cement industry to offset combustion of purchased fuels like coal, oil or natural gas. CO₂ emissions associated with the stationary combustion of fossil-based waste fuels are included in the National Inventory Report where data are available. Fuel use data reported by the cement industry, using CO₂ accounting and reporting standards developed by the World Business Council for Sustainable Development (WBSCD, 2005), were used to generate the emission factors in [Table A6.1–14](#) from 1990 to 2016. From 2017 onwards, fuel use data and emissions were reported by cement facilities through the Greenhouse Gas Reporting Program (GHGRP) (ECCC, 2024a). An annual weighted average EF was developed based on reported data. CO₂ emissions associated with the stationary combustion of biogenic waste fuels are reported as a memo item in the common reporting tables (CRT) of the Energy sector. See section [A6.6.1](#) for the emission factor associated with biogenic waste fuels.

Some municipal solid wastes and medical wastes are combusted in waste-to-energy facilities. See section [A6.7.2](#) for the emission factors associated with these other fuels.

A6.1.5.2. CH₄

CH₄ emission factors for fossil-based alternative fuels ([Table A6.1–14](#)) were adapted from the 2006 IPCC Guidelines (IPCC, 2006). See section [A6.6.2](#) for the emission factors associated with bio-based alternative fuels.

Some municipal solid wastes and medical wastes are combusted in waste-to-energy facilities. See section [A6.7.2](#) for the emission factors associated with these other fuels.

A6.1.5.3. N₂O

N₂O emission factors for fossil-based alternative fuels ([Table A6.1–14](#)) were adapted from the 2006 IPCC Guidelines (IPCC, 2006). See section [A6.6.3](#) for the emission factors associated with bio-based alternative fuels.

Table A6.1–14 Emission Factors for Alternative Fuels			
Year	Emission Factor (kg/GJ)		
	Cement Industry Waste Fuel		
	CO ₂ ^a	CH ₄ ^b	N ₂ O ^b
1990	78.8	0.03	0.004
1995	77.6	0.03	0.004
2000	77.6	0.03	0.004
2005	81.1	0.03	0.004
2006	79.4	0.03	0.004
2007	80.1	0.03	0.004
2008	79.5	0.03	0.004
2009	78.2	0.03	0.004
2010	79.3	0.03	0.004
2011	79.1	0.03	0.004
2012	79.1	0.03	0.004
2013	79.1	0.03	0.004
2014	79.1	0.03	0.004
2015	79.1	0.03	0.004
2016	79.1	0.03	0.004
2017	88.7	0.03	0.004
2018	83.0	0.03	0.004
2019	88.9	0.03	0.004
2020	68.5	0.03	0.004
2021	81.7	0.03	0.004
2022	75.9	0.03	0.004
2023	77.0	0.03	0.004

Notes:
a. Years 1990–2016 adapted from WBSCD (2005). Years 2017–2023 adapted from ECCC (2024a).
b. Adapted from IPCC (2006)

Some municipal solid wastes and medical wastes are combusted in waste-to-energy facilities. See section [A6.7.2](#) for the emission factors associated with these other fuels.

A6.1.6. Mobile Combustion

A6.1.6.1. CO₂

CO₂ emission factors for mobile combustion are dependent on fuel properties and are generally the same as those used for stationary combustion fuels.

A6.1.6.2. CH₄

Emissions of CH₄ from fuel combustion are technology-dependent. Mode-specific CH₄ emission factors have been developed based on technologies typically used in Canada, and are summarized in [Table A6.1–15](#). The factors were initially adopted from a review of available knowledge and an analysis of combustion technologies. A number of on-road CH₄ emission factors were subsequently refined with updated Canadian and U.S. emissions test results (Environment Canada, 2006, 2009; Graham et al., 2008). Light-duty on-road trucks and vehicles of model year 2017 and newer are associated with Tier 3 emissions standards. CH₄ emission factors for Tier 3 vehicles were established using MOVES3-based results (Oak Leaf Environmental Inc., 2022). The CH₄ emission factor applicable to off-road gasoline two-stroke engines was parsed into gasoline and lubricating oil components based on relative proportions of energy in the gasoline-lubricating oil blend (ECCC, 2022c).

Over 50 aircraft-specific aviation turbo fuel CH₄ emission factors from the 2006 IPCC Guidelines (IPCC, 2006) are used in the Tier 3 civil aviation model (Aviation Greenhouse Gas Emission Model—AGEM). [Table A6.1–15](#) displays a national overall average implied emission factor (refer to section [A3.4.2.3](#) for more information on AGEM).

A6.1.6.3. N₂O

Emissions of N₂O from fuel combustion are technology-dependent. Mode-specific N₂O emission factors have been developed based on technologies typically used in Canada. The factors were initially adopted from a review of available knowledge and an analysis of combustion technologies. Several on-road N₂O emission factors were subsequently refined with updated Canadian and U.S. emissions test results (Environment Canada, 2006, 2009; Graham et al., 2008, 2009). Light-duty on-road trucks and vehicles of model year 2017 and newer are associated with Tier 3 emissions standards. N₂O emission factors for Tier 3 vehicles were established using certification-based results (Oak Leaf Environmental Inc., 2022).

The N₂O emission factor applicable to off-road gasoline two-stroke engines was parsed into gasoline and lubricating oil components based on relative proportions of energy in the gasoline-lubricating oil blend (ECCC, 2022c).

In particular, the updated test data highlighted the effect of high-sulphur gasoline on N₂O emission factors. Vehicles fuelled with high-sulphur gasoline for most of their useful lives generally emitted higher levels of N₂O than those run on low-sulphur gasoline (Environment Canada, 2009).

Table A6.1–15 Emission Factors for Energy Mobile Combustion Sources

Mode†	Emission Factors (g/L fuel)		
	CO ₂	CH ₄	N ₂ O
Road Transport			
Gasoline Vehicles			
Light-duty Gasoline Vehicles (LDGVs)			
Tier 3	2 307.3 ^a	0.111 ^r	0.007 ^r
Tier 2	2 307.3 ^a	0.14 ^c	0.022 ^d
Tier 1	2 307.3 ^a	0.23 ^e	0.47 ^e
Tier 0	2 307.3 ^a	0.32 ^f	0.66 ^g
Oxidation Catalyst	2 307.3 ^a	0.52 ^h	0.20 ^f
Non-catalytic Controlled	2 307.3 ^a	0.46 ^h	0.028 ^f
Light-duty Gasoline Trucks (LDGTs)			
Tier 3	2 307.3 ^a	0.111 ^r	0.007 ^r
Tier 2	2 307.3 ^a	0.14 ^c	0.022 ^d
Tier 1	2 307.3 ^a	0.24 ^e	0.58 ^e
Tier 0	2 307.3 ^a	0.21 ^h	0.66 ^g
Oxidation Catalyst	2 307.3 ^a	0.43 ^h	0.20 ^f
Non-catalytic Controlled	2 307.3 ^a	0.56 ^f	0.028 ^f
Heavy-duty Gasoline Vehicles (HDGVs)			
Three-way Catalyst	2 307.3 ^a	0.068 ^h	0.20 ^h
Non-catalytic Controlled	2 307.3 ^a	0.29 ^f	0.047 ^f
Uncontrolled	2 307.3 ^a	0.49 ^f	0.084 ^f
Motorcycles			
Non-catalytic Controlled	2 307.3 ^a	0.77 ^c	0.041 ^c
Uncontrolled	2 307.3 ^a	2.3 ^f	0.048 ^f
Diesel Vehicles			
Light-duty Diesel Vehicles (LDDVs)			
Advanced Control*	2 680.50 ^a	0.051 ^f	0.22 ^f
Moderate Control	2 680.50 ^a	0.068 ^f	0.21 ^f
Uncontrolled	2 680.50 ^a	0.10 ^f	0.16 ^f
Light-duty Diesel Trucks (LDDTs)			
Advanced Control*	2 680.50 ^a	0.068 ^f	0.22 ^f
Moderate Control	2 680.50 ^a	0.068 ^f	0.21 ^f
Uncontrolled	2 680.50 ^a	0.085 ^f	0.16 ^f
Heavy-duty Diesel Vehicles (HDDVs)			
Advanced Control	2 680.50 ^a	0.11 ⁱ	0.151 ⁱ
Moderate Control	2 680.50 ^a	0.14 ^f	0.082 ^f
Uncontrolled	2 680.50 ^a	0.15 ^f	0.075 ^f
Natural Gas Vehicles	1.9 ^b	0.0088 ^f	0.00006 ^f
Propane Vehicles	1 515 ^b	0.64 ^f	0.028 ^f
Off-road			
Off-road Gasoline 2-stroke	2 307.3 ^a	10.56 ^q	0.013 ^q
Off-road Gasoline 4-stroke	2 307.3 ^a	5.08 ^l	0.064 ^m
Off-road Diesel < 19kW	2 680.50 ^a	0.073 ^l	0.022 ^l
Off-road Diesel ≥ 19kW, Tier 1 - 3	2 680.50 ^a	0.073 ^l	0.022 ^l
Off-road Diesel ≥ 19kW, Tier 4	2 680.50 ^a	0.073 ^l	0.227 ^l
Off-road Lubricating Oil 2-stroke	2 705.0 ^q	12.69 ^q	0.016 ^q
Off-road Natural Gas	1.9 ^b	0.0088 ^f	0.00006 ^f
Off-road Propane	1 515 ^b	0.64 ^f	0.087 ^l
Railways			
Diesel Train	2 680.50 ^a	0.149 ^m	1.029 ^m
Marine			
Gasoline	2 307.3 ^a	0.21931 ^m	0.06266 ^m
Diesel	2 680.50 ^a	0.25193 ^m	0.07198 ^m
Light Fuel Oil	2 753 ^b	0.2555 ^m	0.073 ^m
Heavy Fuel Oil	3 156 ^b	0.2856 ^m	0.0816 ^m
Kerosene	2 559.70 ⁿ	0.2471 ^m	0.0706 ^m
Aviation			
Aviation Gasoline	2 325.40 ^j	2.19 ^j	0.23 ^j
Aviation Turbo Fuel	2 559.70 ^b	0.018 ^k	0.0711 ^m
Renewable Fuels			
Ethanol	1 508.04 ^{a, o}	**	**
Biodiesel	2 472.2 ^{a, o, p}	***	***

Notes:

† In the context of Transportation Modes, Tiers refer to increasingly stringent emission standards, enabled through advancements in emission control technologies. It should not be confused with IPCC GHG estimation methodologies.

* Advanced control diesel emission factors are used for Tier 2 diesel vehicle populations.

** Gasoline CH₄ and N₂O emission factors (by mode and technology) are used for ethanol.

*** Diesel CH₄ and N₂O emission factors (by mode and technology) are used for biodiesel.

a. ECCC (2017b)

b. McCann (2000)

c. Adapted from Environment Canada (2006)

d. Adapted from Environment Canada (2006) and Graham et al. (2009)

e. Adapted from Environment Canada (2009)

f. SGA Energy (2000)

g. Adapted from Barton & Simpson (1994)

h. ICF Consulting (2004)

i. Graham et al. (2008)

j. Jaques (1992)

k. National overall average emission factor for the whole time series based on 2006 IPCC Guidelines (IPCC 2006). Refer to section A3.1.4.2.3 of Annex 3.1 for further information.

l. Oak Leaf Environmental Inc (2017)

m. IPCC (2006) Converted into g/L using gross calorific value where necessary.

n. Assumed McCann (2000) aviation turbo-fuel emission factor.

o. Refer to section 3.5.1 Chapter 3 for further information.

p. BioMer (2005)

q. ECCC (2022)

r. Oak Leaf Environmental Inc. (2022)

A6.2. Industrial Processes

A6.2.1. Mineral Products

Emission factors used to estimate emissions from the production and use of mineral products are listed in [Table A6.2–1](#).

Table A6.2–1 Range of CO ₂ Emission Factors for Mineral Products		
Category	Mineral Product	Emission Factor (g CO ₂ / kg of mineral product)
Cement Production	Clinker	515–533 ^a
	Total Organic Carbon (TOC)	9.54–13.1 ^a
Lime Production	High-Calcium Lime	751–759 ^b
	Dolomitic lime	865–889 ^b
Other Limestone and Dolomite Use	Limestone	418–453 ^c
	Dolomite	445–468 ^c
Other Uses of Soda Ash	Soda Ash	415 ^c
Non-Metallurgical Magnesia Production	Magnesite	522 ^c
Notes: a. Range of year-specific emission factors provided by the Cement Association of Canada (2015) and ECCC (2024a). b. Range of year-specific emission factors developed based on information from the Canadian Lime Institute (CLI) (2008) and ECCC (2024a). c. Range of year-specific emission factors from AMEC Earth & Environmental (2006) and ECCC (2024a).		

A6.2.2. Chemical Industry

[Table A6.2–2](#), [Table A6.2–3](#), [Table A6.2–4](#) and [Table A6.2–5](#) present the emission factors used for categories included under the Chemical Industry subsector, as well as the sources from which these factors were obtained.

Table A6.2–2 Emission Factors for Ammonia Production			
	Average Ammonia-to-Feed Fuel Factor ^a m ³ natural gas / tonne of NH ₃	Emission Factor g CO ₂ / m ³ of natural gas	Emission Recovery Factor g CO ₂ / kg of urea
Ammonia Production	671	Marketable natural gas emission factors found in NIR Table A6.1–1 are used.	733
Note: a. Facility-specific fuel factors are used and these are confidential.			

Table A6.2–3 N ₂ O Emission Factors for Nitric Acid and Adipic Acid Production		
Category	Process Description	N ₂ O Emission Factor (kg N ₂ O / t product)
Nitric Acid Production ^a	Sector-wide weighted average in 2023	0.561
Adipic Acid Production	Oxidation reaction of cyclohexanone and cyclohexanol mixture without N ₂ O abatement	300 ^b
Notes: a. Emission factors for nitric acid production process types and abatement technologies are confidential, or could expose confidential activity data. b. IPCC (2000)		

Table A6.2–4 **Emission Factors for Petrochemical Products**

Petrochemical Product	Emission Factor	Type
Silicon Carbide	11.6 kg CH ₄ / t (tonne) product	IPCC default ^a
Calcium Carbide	4.8 kg CH ₄ / t product	Derived from CH ₄ emission factor for silicon carbide and the ratio of IPCC default Calcium Carbide CO ₂ emission factor to IPCC default Silicon Carbide CO ₂ emission factor (i.e. 11.6 (kg CH ₄ / t SiC) * (1.09 tCO ₂ / tCaC ₂ / 2.62 tCO ₂ / tSiC))
Carbon Black	1.29 kg CH ₄ / t product	Sector-wide weighted average ^b
	0.032 kg N ₂ O / t product	Sector-wide weighted average ^b
Ethylene	0.039 kg CH ₄ / t product	Sector-wide weighted average ^b
	0.0055 kg N ₂ O / t product	Sector-wide weighted average ^b
	0.411 t CO ₂ / t product	Sector-wide weighted average ^c
Ethylene Dichloride	0.0142 kg CH ₄ / t product	IPCC default ^d
Ethylene Oxide	0.5202 t CO ₂ / t product	Sector-wide average ^b
	1.79 kg CH ₄ / t product	IPCC default ^d
Styrene	4 kg CH ₄ / t product	IPCC default ^a
Methanol	0.105 kg CH ₄ / t product	Average of sector-wide weighted datab (1990–2006) and facility reported data (2011–2021)
	0.010 kg N ₂ O / t product	Average of sector-wide weighted datab (1990–2006) and facility reported data (2011–2021)
	0.754 t CO ₂ / t product	Average of sector-wide weighted datac (1990–2006) and facility reported data (2011–2021)
Other Uses of Urea	0.733 t CO ₂ / t product use	IPCC default ^d

Notes:

a. Default value from IPCC/OECD/IEA (1997)

b. Cheminfo Services (2010)

c. Cheminfo Services (2015); emission factors may vary if changes are made to the composition of feed.

d. IPCC (2006)

Table A6.2–5 **Emission Factor for By-Product Emissions from Fluorochemical Production**

Process	Emission Factor
HCFC-22 production	0.04 t HFC-23 emitted / t HCFC-22 produced ^a

Note:

a. IPCC (2006)

A6.2.3. Metal Production

The range of the metallurgical coke emission factors and other parameters used for estimating emissions from Iron and Steel Production are found in [Table A6.2–6](#), and [Table A6.2–7](#).

Table A6.2–6 **Range of CO₂ Emission Factors for the Iron and Steel Industry**

Parameter	Emission Factor	Unit
Iron ore reduction with coke	3.1–3.3 ^a	t CO ₂ / t (tonne) coke used
Electrode consumption in electric arc furnaces	4.5–7.7 ^b	kg CO ₂ / t steel
Emissions from basic oxide furnaces	0.14–0.23 ^b	kg CO ₂ / t steel
Limestone use	418–453 ^c	CO ₂ / kg CaCO ₃
Dolomite use	445–468 ^c	CO ₂ / kg MgCO ₃

Notes:

a. Range of year-specific emission factors provided in Cheminfo Services (2010) and ECCC (2024a).

b. Provided by the Canadian Steel Producers Association and ECCC (2022). Chan K. (2009). Personal communication (email from Chan K to Pagé M, Environment Canada, dated July 21, 2009). Canadian Steel Producers Association.

c. Range of year-specific emission factors from AMEC Earth & Environmental (2006) and ECCC (2024a).

Tier 1-type emission factors for the category of Aluminium Production and the sources from which these emission factors were obtained are shown in [Table A6.2–8](#).

Tier 1 CO₂ emission factors for the Magnesium Production category and shown in [Table A6.2–9](#).

Table A6.2–7 Range of Carbon Contents for the Iron and Steel Industry	
Parameter	Carbon Contents (%) ^a
Pig iron (production of pig iron) from BF's	4.4–4.7
Pig iron (includes hot metal, cold iron, DRI and pig iron) for steel making	3.4–3.9
Crude steel produced in BOF	0.1–0.13
Crude steel produced in EAF	0.14–0.26
Scrap steel	0.1–1.2
Note: a. Range of values from CSPA (2009) and ECCC (2024a)	

Table A6.2–8 Tier 1 Emission Factors for Aluminium Production			
Cell Technology Type	Emission Factors (kg / t product)		
	CO ₂	Carbon Tetrafluoride (CF ₄)	Carbon Hexafluoride (C ₂ F ₆)
Side-worked pre-baked	1 600	1.6	0.4
Centre-worked pre-baked	1 600	0.4	0.04
Horizontal stud Söderberg	1 700	0.4	0.03
Vertical stud Söderberg	1 700	0.8	0.04
Note: Data Source: International Aluminium Institute (IAI) (2006)			

Table A6.2–9 Tier 1 CO ₂ Emission Factors for Magnesium Production	
Carbonate Raw-Material used in Mg Production	Emission Factor (tonnes CO ₂ emission/tonne primary Mg produced) ^a
Dolomite	5.13
Magnesite	2.83
Note: a. IPCC (2006)	

A6.2.4. Non-Energy Products from Fuels and Solvent Use

The use of fossil fuels as feedstock or for other non-energy use (NEU) may result in emissions during the life of manufactured products. To estimate CO₂ emissions from NEU of natural gas, an emission factor of 38 g CO₂/m³ was used. This emission factor excludes the feedstock use of natural gas to produce ammonia, and it is derived from the NEU of natural gas data found in the 2005 Cheminfo Study (Cheminfo Services, 2005).

[Table A6.2–10](#) shows the emission factors used to develop CO₂ emission estimates for non-energy applications of natural gas liquids and petroleum products, respectively. The emission factors for NEU petroleum coke are found in [Table A6.1–7](#). The 2011 emission factor value for Upgrading Facilities in [Table A6.1–7](#) has been used for Ontario across the time series. For the other provinces, the 2011 emission factor value for Refineries and Others is used across the time series. The emission factors associated with NEU of coal are referenced in [Table A6.1–10](#).

Table A6.2–10 CO₂ Emission Factors for Non-Energy Use of Natural Gas Liquids and Petroleum Products

Product	Fraction of Carbon Stored in Product	CO ₂ Emission Factor (g CO ₂ / L)
Natural Gas Liquids		
Propane	0.8 ^a	303 ^b
Butane	0.8 ^a	349 ^b
Ethane	0.8 ^a	197 ^b
Petroleum Products		
Petrochemical Feedstocks ^c	0.8 ^a	500 ^d
Napthas ^e	0.75 ^a	625 ^d
Lubricating Oils and Greases ^f	0.2 ^g	2 260 ^d
Petroleum Used for Other Products ^h	0.5 ^a	1 450 ^d

Notes:

a. IPCC/OECD/IEA (1997)

b. McCann (2000)

c. Carbon factor for Petrochemical Feedstocks is 680 g of carbon per litre (C/L) (Jaques 1992).

d. The resulting CO₂ emission factor is calculated by multiplying the carbon factor for each product by the molecular weight ratio between CO₂ and carbon (44/12) and by (1-fraction of carbon stored in product).

e. Carbon factor for Napthas is 680 g C/L (Jaques 1992).

f. Carbon factor for Lubricating Oils and Greases is 770 g C/L (Jaques 1992).

g. IPCC (2006)

h. Carbon factor for Petroleum Used in Other Products is 790 g C/L (Jaques 1992).

A6.2.5. Electronics Industry

The use of perfluorocarbons (PFCs), sulphur hexafluoride (SF₆) and nitrogen trifluoride (NF₃) in integrated circuit or semiconductor manufacturing, electrical environmental testing, gross leak testing and thermal shock testing creates GHG emissions of their respective source gases. The use of PFCs and NF₃ in the integrated circuit or semiconductor manufacturing industry can also lead to by-product PFC emissions. The emission factors for the use of PFCs, SF₆ and NF₃ in the electronics industry are summarized in [Table A6.2–11](#).

Table A6.2–11 Emission Factors for the use of PFCs, SF₆ and NF₃ in the Electronics Industry

Application	GHG Source	IPCC Tier	Emission Rate (%) ^a	By-Product Emission Rate
Integrated Circuit or Semiconductor Manufacturing ^b	CF ₄	T2A	90	N/A
	CF ₄	T2B – CVD	90	N/A
	CF ₄	T2B – Etching	70	N/A
	C ₂ F ₆	T2A	60	0.2 kg CF ₄ / kg C ₂ F ₆
	C ₂ F ₆	T2B – CVD	60	0.1 kg CF ₄ / kg C ₂ F ₆
	C ₂ F ₆	T2B – Etching	40	0.4 kg CF ₄ / kg C ₂ F ₆
	c-C ₄ F ₈	T2A	10	0.1 kg CF ₄ / kg c-C ₄ F ₈ , 0.1 kg C ₂ F ₆ / kg c-C ₄ F ₈
	c-C ₄ F ₈	T2B – Etching	20	0.2 kg CF ₄ / kg c-C ₄ F ₈ , 0.2 kg C ₂ F ₆ / kg c-C ₄ F ₈
	SF ₆	T2A	20	N/A
	NF ₃	T2A	20	0.09 kg CF ₄ / kg NF ₃
Other Emissive Applications	NF ₃	T2B – Etching	20	N/A
	PFCs	T2	50% first year / 50% second year	N/A

Notes:

N/A = Not available

a. IPCC (2006)

b. When available, confidential company/gas/process-specific values are used for the fraction of gas volume fed into process types with emission control technology and the fraction of gas destroyed by the emission control technology (respectively, ai and di in the IPCC Guidelines).

A6.2.6. Product Uses as Substitutes for Ozone Depleting Substances

The use of halocarbons in various applications, such as air conditioning (AC), refrigeration, aerosols, foam blowing, solvents and fire extinguishing, result in hydrofluorocarbon (HFC) and PFC emissions.

Table A6.2–12 and Table A6.2–13 summarize emission rates used to estimate HFC and PFC emissions, respectively.

Table A6.2–12 HFCs as Ozone Depleting Substance (ODS) Substitutes – Assembly, In-Service and End-of-Life Emission Factors (%)				
Application/Sub-Application	Assembly ^a	In-Service ^b	End-of-Life ^c	Life Time (years)
Aerosols^d	-	50% first year / 50% second year	-	2
Blowing agent in foams^d				
Open-cell foam	100	-	-	-
Closed-cell foam	10	4.5	100	23
Air conditioning (equipment manufactured in Canada)^e				
Air conditioner units in motor vehicles	0.5	10	75	13
Chillers (specify centrifugal or reciprocating)	1	4.7	5	17
Residential (air conditioners, dehumidifiers, etc.)	1	4	20	17
Air conditioning (manufactured elsewhere)^e				
Air conditioner units in motor vehicles	-	10	75	13
Chillers (specify centrifugal or reciprocating)	-	4.7	5	17
Residential (air conditioners, dehumidifiers, etc.)	-	4	20	17
Refrigeration (equipment manufactured in Canada)^e				
Commercial transport	1	15	30	13
Commercial and institutional (retail foods, vending machines, etc.)	1	10	30	17
Industrial (warehouses, process equipment, etc.)	1	10	30	17
Residential (freezers, refrigerators)	0.6	0.5	30	15
Other equipment (specify)	1.0	10.8	30	15
Refrigeration (manufactured elsewhere)^e				
Commercial transport	-	15	30	13
Commercial and institutional (retail foods, vending machines, etc.)	-	10	30	17
Industrial (warehouses, processes, etc.)	-	10	30	17
Residential (refrigerators, freezers, etc.)	-	0.5	30	15
Other equipment (specify)	-	10.1	30	15
Solvent^d	-	50% first year / 50% second year	-	2
Fire suppression/extinguishing systems^d				
Portable (mobile) systems	-	4	5	18
Total flooding (fixed) systems	-	2	5	18
Miscellaneous^d (Hospital / institutional sterilizing and leak testing)	-	50% first year / 50% second year	-	2
Other (specify)^d	-	50% first year / 50% second year	-	2
Notes: a. Percentage of losses of the HFC charged into new equipment. b. Release percentage of HFC bank (by application) during operation. c. Release percentage of HFC bank (by application) during disposal. d. IPCC (2006) e. Environment and Climate Change Canada (2015)				

Table A6.2–13 PFCs as Ozone Depleting Substance (ODS) Substitutes – Assembly, In-Service and End-of-Life Emission Factors (%)

Application/Sub-Application	Assembly ^a	In-Service ^b	End-of-Life ^c	Life Time (years)
Air Conditioning				
Mobile A/C	0.35	15	100	12
Residential and Commercial A/C, including Heat Pumps	0.6	5.5	100	15
Foam Blowing Agents				
Closed-Cell Foam	45	0.75% of original charge	100 (43.75% of original charge)	15
Refrigeration				
Industrial Refrigeration including Food Processing and Cold Storage	1.75	16	100	15
Medium and Large Commercial Refrigeration	1.75	22.5	100	15
Solvents				
	-	50% first year / 50% second year	-	2

Notes:

a. Percentage of losses of the PFC charged into new equipment.

b. Annual release percentage of PFC bank (by application) during operation.

c. Release percentage of PFC bank (by application) during disposal.

Source: IPCC (2006)

A6.3. Other Product Manufacture and Use

The uses of N₂O as an anaesthetic and as a propellant result in N₂O emissions. The emission factors used are shown in [Table A6.3–1](#).

The use of perfluorocarbons (PFCs) in contained applications (such as use as an electrical insulator or as a dielectric coolant) results in PFC emissions. The emission factors used for contained applications are shown in [Table A6.3–2](#).

The use of urea-based diesel exhaust fluid (DEF) in diesel vehicles equipped with selective catalytic reduction (SCR) systems results in CO₂ emissions, the rate of which is dependent on the purity factor of urea in DEF as well as the dosing rate of urea to diesel consumption as per [Table A6.3–3](#).

Table A6.3–1 Emission Factors for N₂O Usage (Medical and Propellant)

Product	Application	N ₂ O Emission Rate (%)
N ₂ O Use	Anaesthetic Usage	100
	Propellant Usage	100

Note: Source: IPCC (2006)

Table A6.3–2 Emission Factor for PFC Emissions from Other Contained Product Uses

Process	PFC Emissions from Other Contained Sources
Assembly	1% of charge
Annual Leakage Rate	2% of stock
Disposal	100% of remaining stock
Product Lifespan	15 years

Note: Source: IPCC (2000)

Table A6.3–3 Emission Factors for Use of Urea in SCR Vehicles

Product	DEF Purity	Dosing Rate
Urea use in SCR Vehicles	32.50%	2% of diesel consumption

Note: Source: IPCC (2006)

A6.4. Agriculture

The sources of agricultural GHGs are enteric fermentation, manure management, field burning of crop residues, agricultural soils (including nitrous oxide emissions from mineralization/immobilization associated with loss/gain of soil organic matter), and agricultural use of lime, urea and other carbon-containing fertilizers. The most significant sources use country-specific Tier 2 methodologies. Carbon dioxide emissions from liming, urea application and other carbon-containing fertilizers are calculated based on the total quantity of carbon (C) contained in these products. Ammonia emissions from synthetic nitrogen (N) application are estimated using a country-specific modelling method as noted in [Annex 3.4](#). Finally, indirect emissions from ammonia volatilization and nitrogen leaching are calculated based on the IPCC default emission factors provided in [Table A6.4–27](#).

Those emission factors for agriculture calculated based on country-specific methodologies are described in detail in [Annex 3.4](#). In most cases, annual emission factors by gas presented in this annex represent an overall weighted average, which includes a number of underlying factors that influence regional emission rates. Emission estimates are developed using more detailed emission factors, as described in [Annex 3.4](#). For enteric fermentation emissions from cattle, weighted national emission factors and the methodology for generating emission factors are detailed in [section A3.4.2.1](#). In the case of manure management CH₄, the methodology for generating emission factors is described in [A3.4.3](#), and weighted national emission factors are presented in [A3.4.3.5](#). For manure management N₂O emissions, the methodologies for calculating direct and indirect N₂O emissions are described in [sections A3.4.4.1 and A3.4.4.2](#), respectively. Finally, the methodologies for generating N₂O emission factors for direct emissions from agricultural soils and pasture, range and paddock (PRP), are described in [A3.4.5.1](#). Cattle are described using an approach consistent with [common reporting tables \(CRT\)](#).² For enteric fermentation, Dairy Cattle includes only dairy cows, while for manure management and PRP, Dairy Cattle includes dairy cows and dairy heifers.

A compilation of emission factors for agriculture is provided here in [Table A6.4–1](#) to [Table A6.4–29](#).

A6.4.1. Enteric Fermentation

Year	EF _{(EF)T} (kg CH ₄ head ⁻¹ yr ⁻¹) ^a							
	Dairy Cows	Dairy Heifers	Bulls	Beef Cows	Beef Heifers	Heifers for Slaughter ^b	Steers ^b	Calves
1990	115.4	79.4	108.0	105.9	82.5	44.7	41.4	43.8
2005	125.0	77.2	119.9	114.4	87.0	52.8	46.0	43.6
2017	138.1	76.7	130.2	120.9	91.4	53.7	48.4	43.8
2018	139.6	76.7	125.5	120.6	91.4	54.0	48.7	43.9
2019	142.1	76.7	124.2	120.4	91.2	54.2	49.3	43.8
2020	142.9	76.7	124.2	120.5	91.3	54.2	49.6	43.9
2021	145.3	76.7	127.6	120.0	90.9	54.2	49.7	43.9
2022	145.8	76.7	123.2	120.7	91.4	54.6	49.9	43.9
2023	145.6	76.7	124.6	121.5	91.8	54.4	49.6	44.0

Notes:

a. Enteric emission factors are derived from Boadi et al. (2004), modified to take account of trends in milk production in dairy cows and carcass weights for several beef cattle categories.

b. Reported as kg head⁻¹ yr⁻¹; however, emissions are calculated based on time to slaughter.

² Canada's 2023 CRT tables are available online at: <https://unfccc.int/ghg-inventories-annex-i-parties/2023>.

Table A6.4–2 Methane Emission Factors (EF) for Enteric Fermentation for Non-Cattle Animals

Non-cattle Animal Category	Enteric Fermentation EF (kg CH ₄ head ⁻¹ yr ⁻¹)
Pigs	
Boars	1.5
Sows	1.5
Pigs < 20 kg	1.5
Pigs 20–60 kg	1.5
Pigs > 60 kg	1.5
Other Livestock	
Sheep	8
Lambs	8
Goats	5
Horses	18
Bison	55
Llamas & Alpacas	8
Elk & Deer	20
Wild Boars	1.5
Fox	N/A
Mink	N/A
Rabbits	N/A
Mules and Asses	10
Poultry	
Chickens	N/A
Hens	N/A
Turkeys	N/A
Notes: N/A = Not available Data source: IPCC (2006) (Vol. 4: Agriculture, Forestry and Other Land Uses, Table 10.10)	

A6.4.2. Manure Management

Table A6.4–3 Maximum Methane-Producing Potential (B₀) by Animal Category

Animal Category	Maximum CH ₄ -Producing Potential (B ₀) (m ³ kg ⁻¹ VS)
Dairy Cattle ^a	0.24
Non-dairy Cattle ^b	0.19
Sheep	0.19
Goats	0.18
Horses	0.3
Swine	0.48
Hens	0.39
Broilers	0.36
Turkeys	0.36
Notes: VS = Volatile solids a. Dairy cattle include dairy cows and dairy heifers. b. The non-dairy cattle value is also used for bison. Data source: IPCC (2006) (Vol. 4: Agriculture, Forestry and Other Land Uses, Tables 10A-5 to 10A-9)	

Table A6.4–4 **Methane Conversion Factors (MCF) by Animal Category and Manure Management System**

Animal Categories	Liquid Systems (MCF _L)	Solid Storage and Drylot (MCF _{SSD})	Pasture, Range and Paddock (MCF _{PRP})	Other Systems (MCF _O)
Non-dairy Cattle ^a	0.2	0.02	0.01	0.01
Poultry	0.2	0.015	0.015	0.015
Horses	NA	0.01	0.01	0.01
Goats	NA	0.01	0.01	NA
Sheep	0.2	0.01	0.01	0.01
Lambs	0.2	0.01	0.01	0.01

Notes:

NA = Not applicable

a. Non-dairy cattle values are also used for bison.

Source: IPCC (2006) (Vol. 4: Agriculture, Forestry and Other Land Uses, tables 10A-5 to 10A-9 – cool climate, average annual temperature 12°C)

Table A6.4–5 **Methane Conversion Factors (MCF) for Dairy Cattle and Swine**

Manure Management System	Manure Management Subsystem	Crust Formation	MCF
Liquid	Earthen Basin	No crust	0.2
	Earthen Basin	Crust	0.13
	Tank	No crust	0.2
	Tank	Crust	0.13
	Slatted floor	N/A	0.2
Solid	Exercise Yard	N/A	0.01
	Pack	N/A	0.01
	Pile	N/A	0.02
Compost		N/A	0.005
Pasture Range Paddock		N/A	0.01
		N/A	0.01

Notes:

N/A = Not available

Source: IPCC (2006) (Vol. 4: Agriculture, Forestry and Other Land Uses, Table 10.17 – cool climate, average annual temperature 12°C)

Table A6.4–6 **Emission Factors for Estimating CH₄ Emissions from Manure Management for Cattle by Subcategory, from 1990 to 2023**

Year	EF _{(MM)T} (kg CH ₄ head ⁻¹ yr ⁻¹)							
	Dairy Cows	Dairy Heifers ^a	Bulls	Beef Cows	Beef Heifers	Heifers for Slaughter ^b	Steers ^b	Calves
1990	13	8	4.5	4.1	3.2	1.9	1.8	2.2
2005	26	12	4.6	4.3	3.1	2.1	1.9	2.4
2017	38	17	5.0	4.5	3.2	2.1	2.0	2.9
2018	38	17	4.8	4.5	3.2	2.2	2.0	2.9
2019	39	17	4.7	4.5	3.2	2.2	2.0	2.9
2020	39	17	4.7	4.5	3.2	2.2	2.0	2.9
2021	39	17	4.9	4.5	3.2	2.2	2.0	2.9
2022	40	17	4.7	4.5	3.2	2.2	2.0	2.9
2023	40	17	4.7	4.5	3.2	2.2	2.0	2.9

Notes:

a. For dairy heifers, emission factors were estimated using the same B₀, MCF and manure management systems as for dairy cows.b. Reported as kg head⁻¹ yr⁻¹, but emissions are calculated based on time to slaughter.

Table A6.4–7 Emission Factors for Estimating CH₄ Emissions from Manure Management for Swine by Subcategory, from 1990 to 2023

Year	EF _{(MM)T} (kg CH ₄ head ⁻¹ yr ⁻¹)				
	Boars	Sows	Pigs (< 20 kg)	Pigs (20–60 kg)	Pigs (> 60 kg)
1990	7.0	7.3	2.2	4.5	8.3
2005	7.0	7.1	2.2	4.4	8.5
2017	7.0	7.0	2.1	4.3	9.1
2018	7.0	7.0	2.1	4.3	9.1
2019	7.0	7.0	2.2	4.3	9.2
2020	7.0	7.0	2.2	4.3	9.3
2021	7.0	7.0	2.1	4.3	9.4
2022	7.0	7.0	2.2	4.2	9.4
2023	7.0	7.0	2.2	4.3	9.5

Table A6.4–8 2023 CH₄ Emission Factors for Manure Management for All Other Livestock

Non-Cattle Animal Categories	Manure Management Emission Factors EF _(MM) (kg CH ₄ head ⁻¹ yr ⁻¹)
Other livestock	
Sheep	0.33
Lambs	0.22
Goats	0.32
Horses	2.6
Bison	2.1
Elk and deer	0.22
Wild boars ^a	0.56
Foxes	0.68
Mink	0.68
Rabbits	0.08
Mules and asses	0.76
Poultry	
Chickens	0.03
Hens	0.11
Turkeys	0.10
Note: a. Emission factor based on swine VS, assuming 100% solid manure.	

Table A6.4–9 Dairy Cattle and Swine Emission Factors for Manure Nitrogen (N) Lost as N₂O-N by Animal Waste Management Systems

Manure Management System	Manure Management Subsystem	Crust Formation	Emission Factor
Liquid	Earthen Basin	No crust	0
	Earthen Basin	Crust	0.005
	Tank	No crust	0
	Tank	Crust	0.005
	Slatted floor	NA	0.002
Solid	Exercise Yard	NA	0.02
	Pack	NA	0.02
	Pile	NA	0.005
Other	Compost	NA	0.01
Notes: NA = Not applicable Source: IPCC (2006) (Vol. 4: Agriculture, Forestry and Other Land Uses, Table 10.21)			

Table A6.4–10 Emission Factors (EF) for Manure Nitrogen (N) Lost as N₂O-N by Animal Category and Animal Waste Management Systems

	Liquid Systems (EF _L)	Solid Storage and Drylot (EF _{SSD})	Other Systems (EF _O)
Non-dairy Cattle	0.001	0.02	0.005
Poultry	0.001	0.02	0.005
Sheep and Lambs	0.001	0.02	0.005
Goats	0.001	0.02	0.005
Horses	0.001	0.02	0.005
Mules and Asses	0.001	0.02	0.005
Buffalo	0.001	0.02	0.005
Note: Source: IPCC (2006) (Vol. 4: Agriculture, Forestry and Other Land Uses, Table 10.21)			

Table A6.4–11 **Emission Factors (EF) for Manure Nitrogen (N) Lost as N₂O During Storage of Cattle and Swine Manure**

Year	EF (g N ₂ O head ⁻¹ year ⁻¹)				
	1990	2005	2015	2020	2023
Cattle^a					
Dairy Cows ^b	1 268	1 128	930	917	902
Beef Cows	862	999	1 093	1 102	1 116
Bulls	1 305	1 495	1 641	1 575	1 579
Dairy Heifers ^b	938	906	745	751	751
Beef Heifers	680	769	838	843	853
Heifers for Slaughter	320	425	458	467	466
Steers	336	426	468	481	487
Calves	382	383	382	382	383
Swine^c					
Sows	74	29	24	24	24
Boars	95	58	58	57	58
Pigs (<20 kg)	6	2	2	2	2
Pigs (20–60 kg)	26	12	10	9	9
Pigs (>60 kg)	52	25	22	22	23

Notes:

- a. Emission factors are derived from information in Boadi et al. (2004), Marinier et al. (2004) and (2005), and default factors in the 2006 IPCC Guidelines. Derivation of the Tier 2 emission factors is explained in NIR Annex 3.4.
- b. For dairy cows and heifers, nitrogen excretion rates are derived from feed intake information from Lactanet, and manure storage practices are taken from farm management surveys, as described in NIR Annex 3.4.
- c. For swine, nitrogen excretion rates are calculated using default IPCC parameters and country-specific animal mass time series, and manure storage practices are taken from farm management surveys, as described in NIR Annex 3.4.

Table A6.4–12 **2023 Emission Factors (EF) for Manure Nitrogen (N) Lost as N₂O During Storage of Non-Cattle and Non-Swine Manure**

Livestock Category	Emission Factors (EF) ^a (g N ₂ O head ⁻¹ year ⁻¹)
Poultry	
Turkey	54
Hens	13
Pullets	6
Broiler	11
Other Livestock	
Sheep	45
Goat	135
Buffalo	990
Horse	486
Llama and alpacas	163
Lamb	41
Deer	222
Elk	210
Wild boars	280
Rabbit	149
Mink	95
Fox	250
Mules and Asses	265

Note:

- a. Emission factors are derived from information in Marinier et al. (2004) and (2005), and default factors in the 2006 IPCC Guidelines. Derivation of the Tier 2 emission factors is explained in NIR Annex 3.4.

Table A6.4–13 **Emission Factors (EF) for Cattle and Swine Manure Nitrogen (N) Lost Indirectly as N₂O Due to Volatilization and Leaching During Storage**

	EF (g N ₂ O head ⁻¹ yr ⁻¹)				
	1990	2005	2015	2020	2023
Volatilization^a					
Dairy Cow	261	263	217	214	209
Beef Cow	98	105	113	112	113
Bull	162	162	173	162	160
Dairy Heifers	197	187	152	151	150
Beef Heifers	86	83	88	87	88
Heifers for Slaughter	48	48	54	54	53
Steer	49	58	59	59	61
Calf	46	43	42	41	41
Sow	63	58	49	48	48
Boar	60	54	45	42	43
Pig (<20 kg)	5	5	4	4	4
Pig (20–60 kg)	21	20	17	17	17
Pig (>60 kg)	43	42	38	40	41
Leaching^b					
Dairy Cow	23	15	10	9	9
Beef Cow	0	0	0	0	0
Bull	0	0	0	0	0
Dairy Heifers	16	12	9	9	9
Beef Heifers	0	0	0	0	0
Heifers for Slaughter	0	0	0	0	0
Steer	0	0	0	0	0
Calf	0	0	0	0	0
Sow	1.2	0.2	0.1	0.1	0.1
Boar	1.6	0.7	0.7	0.7	0.7
Pig (<20 kg)	0.10	0.02	0.01	0.01	0.01
Pig (20–60 kg)	0.5	0.1	0.1	0.1	0.1
Pig (>60 kg)	0.9	0.3	0.2	0.2	0.2

Notes:

- a. Indirect N₂O from volatilization is calculated using the climate-specific EF₄ values from Table 11.3 in Volume 4, Chapter 11 of the 2019 Refinement to the 2006 IPCC Guidelines. Volatilization is calculated based on Sheppard et al. (2010), Sheppard et al. (2011b) and Chai et al. (2016). Derivation of the Tier 2 emission factors is explained in NIR Annex 3.4.
- b. A Tier 2 method for the calculation of swine and dairy cattle leaching is based on Sheppard et al. (2010), Sheppard et al. (2011b) and Chai et al. (2016). Derivation of the Tier 2 emission factors is explained in NIR Annex 3.4.

Table A6.4–14 Annual Emission Factors (EF) for Cattle and Swine Manure Nitrogen (N) Lost as NH₃ Due to Volatilization During Storage

	EF (kg NH ₃ head ⁻¹ year ⁻¹)				
	1990	2005	2015	2020	2023
Cattle					
Dairy Cow	22	17	13	13	13
Beef Cow	39	67	50	51	49
Bull	4	5	4	4	4
Dairy Heifers	8	6	4	4	4
Beef Heifers	5.7	7	6	7	6
Heifers for Slaughter	2.3	6.1	5.2	4.7	4.8
Steer	4.7	7.6	8.5	8.4	8.8
Calf	17.6	25.7	18.3	18.3	17.6
Swine					
Sow	4.6	7.1	4.7	4.8	4.8
Boar	0.3	0.2	0.1	0.1	0.1
Pig (<20 kg)	1.1	1.7	1.6	1.7	1.7
Pig (20-60 kg)	4.7	7.0	3.7	3.8	3.8
Pig (>60 kg)	9.0	13.6	12.4	13.7	13.7

Note: Volatilization is calculated based on Sheppard et al. (2010), Sheppard et al. (2011b) and Chai et al. (2016). Derivation of the emission factors is explained in NIR Annex 3.4.

Table A6.4–15 2023 Emission Factors (EF) for Manure Nitrogen (N) Lost Indirectly as N₂O Due to Volatilization and Leaching During Storage

Livestock Category	Volatilization Emission Factor (EF) ^a (g N ₂ O head ⁻¹ yr ⁻¹)	Leaching Emission Factor (EF) ^b (g N ₂ O head ⁻¹ yr ⁻¹)
Poultry		
Turkey	16	0
Hens	4	0
Pullets	2	0
Broiler	3	0
Other Livestock		
Sheep	3	0
Goat	10	0
Buffalo	85	0
Horse	29	0
Llama and alpacas	12	0
Lamb	3	0
Deer	41	0
Elk	20	0
Wild boars	46	0
Rabbit	31	0
Mink	8	0
Fox	21	0
Mules and Asses	16	0

Notes:

a. Indirect N₂O from volatilization is calculated using the climate-specific EF₄ values from Table 11.3 in Volume 4, Chapter 11 of the 2019 Refinement to the 2006 IPCC Guidelines. Derivation of the volatilization emission factors is explained in NIR Annex 3.4.

b. Leaching is not calculated as there are no Tier 1 leaching factors available in the 2006 IPCC Guidelines.

Table A6.4–16 2023 Emission Factors (EF) for Manure Nitrogen (N) Lost as NH₃ Due to Volatilization During Storage

Livestock Category	Emission Factor (EF) (kg NH ₃ head ⁻¹ yr ⁻¹)
Poultry	
Turkey	1.0
Hens	0.3
Pullets	0.1
Broiler	0.2
Other Livestock	
Sheep	0.2
Goat	0.6
Buffalo	12
Horse	2.4
Llama and alpacas	0.8
Lamb	0.2
Deer	2.6
Elk	2.6
Wild boars	3.2
Rabbit	1.7
Mink	0.4
Fox	1.2
Mules and Asses	1.3

Note: Volatilization factors are taken from default parameters in the 2006 IPCC Guidelines. Derivation of the emission factors is explained in NIR Annex 3.4.

A6.4.3. Pasture, Range and Paddock

Table A6.4–17 Emission Factors (EF) for Manure Nitrogen (N) Lost as N₂O From Deposition of Cattle Manure on Pasture, Range and Paddock

Year	EF (g N ₂ O kg ⁻¹ N yr ⁻¹) ^{a, b, c, d}									
	AB	BC	MB	NB	NL	NS	ON	PE	QC	SK
1990	0.68	0.68	0.68	8.8	9.5	8.8	9.7	7.5	9.4	0.68
2005	0.68	0.68	0.68	8.8	9.6	8.8	9.7	7.6	9.4	0.68
2020	0.68	0.68	0.68	8.8	9.6	8.9	9.6	7.5	9.4	0.68
2023	0.68	0.68	0.68	8.7	9.5	8.8	9.6	7.5	9.4	0.68

Notes:

- a. Emission factors are derived from Rochette et al. (2014) for eastern Canada, and Lemke et al. (2012) for western Canada.
b. The proportion of excreted manure deposited on pasture is taken from Marinier et al. (2005), for all livestock except dairy cows and heifers.
c. The proportion of excreted manure deposited on pasture by dairy cows and heifers is based on a farm size relationship derived from Sheppard et al. (2011a), as described in NIR Annex 3.4.
d. Derivation of the Tier 2 emission factors is explained in NIR Annex 3.4.

Table A6.4–18 Emission Factors (EF) for Manure Nitrogen (N) Lost Indirectly as N₂O Due to Volatilization and Leaching of Manure Deposited on Pasture, Range and Paddock

	EF (g N ₂ O kg ⁻¹ N yr ⁻¹)									
	AB	BC	MB	NB	NL	NS	ON	PE	QC	SK
Volatilization^a										
1990	1.6	1.8	1.6	2.9	2.3	2.9	3.0	2.7	2.3	1.6
2005	1.6	1.8	1.6	3.1	2.2	3.2	3.4	3.1	2.8	1.6
2015	1.6	1.7	1.6	3.0	1.9	3.0	3.3	2.7	2.6	1.6
2020	1.6	1.7	1.6	2.5	2.2	2.8	3.2	2.6	2.3	1.6
2023	1.6	1.6	1.6	2.8	2.1	2.9	3.2	2.7	2.4	1.6
Leaching^b										
1990	1.9	1.7	2.1	3.5	3.5	3.5	3.1	3.5	3.4	1.6
2005	1.8	1.7	2.1	3.5	3.5	3.5	3.1	3.5	3.4	1.6
2015	1.8	1.7	2.1	3.5	3.5	3.5	3.1	3.5	3.4	1.6
2020	1.8	1.7	2.1	3.2	3.6	3.5	3.0	3.5	3.4	1.5
2023	1.8	1.7	2.1	3.5	3.5	3.5	3.1	3.5	3.4	1.5

Notes:

a. For dairy cattle, volatilization is calculated based on Sheppard et al. (2011b) and Chai et al. (2016), and the default climate-specific indirect N₂O emission factors from the 2019 Refinement to the 2006 IPCC Guidelines are used. For all other livestock the IPCC Tier 1 methodology is used to estimate indirect N₂O emission factors from volatilization. Default parameters are used from the 2006 IPCC Guidelines as described in NIR Annex 3.4.

b. A modified IPCC Tier 1 methodology is used to estimate N₂O emissions from leaching in agricultural soils, as described in NIR Annex 3.4.

Table A6.4–19 Emission Factors (EF) for Manure Nitrogen (N) Lost Indirectly as NH₃ Due to Volatilization of Manure Deposited on Pasture, Range and Paddock

Year	EF (kg NH ₃ kg ⁻¹ N yr ⁻¹)									
	AB	BC	MB	NB	NL	NS	ON	PE	QC	SK
1990	0.20	0.18	0.19	0.13	0.10	0.13	0.14	0.12	0.10	0.20
2005	0.21	0.19	0.20	0.14	0.10	0.14	0.16	0.14	0.13	0.21
2020	0.21	0.19	0.20	0.13	0.09	0.13	0.15	0.12	0.11	0.21
2023	0.21	0.19	0.20	0.13	0.10	0.13	0.15	0.12	0.11	0.21

Note: For dairy cattle, volatilization is calculated based on Sheppard et al. (2011b) and Chai et al. (2016). For all livestock except dairy cattle, the IPCC Tier 1 methodology is used to estimate volatilization. Further detail can be found in NIR Annex 3.4.

Table A6.4–20 Emission Factors (EF) for Crop Residue, Organic and Inorganic Fertilizer Nitrogen (N) Lost as N₂O Following Application to Agricultural Soils

	EF (g N ₂ O kg ⁻¹ N yr ⁻¹)						
	AB	BC	MB	ATL	ON	QC	SK
Organic N							
1990	8.1	6.9	9.0	7.6	15	17	5.0
2005	7.5	7.2	8.4	8.2	16	20	4.2
2015	7.8	8.2	8.8	9.0	16	22	4.1
2020	8.1	8.2	9.6	9.2	16	22	4.0
2021	8.1	8.2	9.7	9.2	16	22	4.1
2022	8.2	8.2	9.9	9.2	16	22	4.0
2023	8.1	8.1	9.9	9.4	16	22	4.0
Inorganic N							
1990	10	5.6	11	12	19	24	7.8
2005	8.7	4.8	11	13	19	28	6.8
2015	8.8	5.1	11	14	20	30	6.5
2020	8.9	4.9	10	14	20	30	6.5
2021	8.9	4.7	10	14	20	29	6.5
2022	9.0	4.7	10	14	20	29	6.5
2023	9.0	4.7	10	14	20	29	6.5
Crop Residue N							
1990	8.2	4.8	8.6	8.5	13	15	6.5
2005	8.0	5.3	8.1	9.2	15	21	5.7
2015	8.2	6.0	8.9	11	16	23	5.6
2020	8.4	5.0	8.7	11	17	25	5.6
2021	8.4	5.7	8.7	11	17	23	5.6
2022	8.4	5.2	8.5	11	17	23	5.6
2023	8.5	5.7	8.6	11	17	24	5.6
Note: Country-specific Tier 2 soil N ₂ O emission factors are calculated as described in NIR Annex 3.4.							

Table A6.4–21 Emission Factors (EF) for Manure Nitrogen (N) Lost as NH₃ from Agricultural Soils

Year	EF (g NH ₃ kg ⁻¹ N yr ⁻¹)									
	AB	BC	MB	NB	NL	NS	ON	PE	QC	SK
1990	235	214	228	295	266	294	254	293	267	239
2005	236	204	196	264	253	258	242	277	256	236
2020	235	199	181	250	250	249	237	265	252	236
2023	235	199	179	251	250	245	236	266	252	235
Notes: For dairy cattle and swine, volatilization is calculated based on Sheppard et al. (2010), Sheppard et al. (2011b) and Chai et al. (2016). For all other livestock the IPCC Tier 1 methodology is used to estimate volatilization. Further detail can be found in NIR Annex 3.4.										

Table A6.4–22 Emission Factors (EF) for Manure Nitrogen (N) Lost Indirectly as N₂O Due to Volatilization and Leaching of Manure Applied to Agricultural Soils

	EF (g N ₂ O kg ⁻¹ N applied yr ⁻¹)									
	AB	BC	MB	NB	NL	NS	ON	PE	QC	SK
Volatilization^a										
1990	1.5	2.7	1.6	5.3	4.8	5.3	5.3	4.6	4.8	1.5
2005	1.5	2.4	1.3	4.8	4.6	4.7	5.0	4.4	4.6	1.5
2020	1.5	2.4	1.2	4.5	4.5	4.5	4.8	4.3	4.6	1.5
2023	1.5	2.4	1.2	4.6	4.5	4.4	4.8	4.3	4.6	1.5
Leaching^b										
1990	1.9	2.3	2.1	3.5	3.5	3.5	3.5	3.1	3.5	1.6
2005	1.8	2.3	2.1	3.5	3.5	3.5	3.5	3.1	3.5	1.6
2020	1.8	2.3	2.1	3.5	3.5	3.5	3.5	3.1	3.5	1.5
2023	1.8	2.3	2.1	3.5	3.5	3.5	3.5	3.1	3.5	1.5

Notes:

a. For dairy cattle and swine, volatilization is calculated based on Sheppard et al. (2010), Sheppard et al. (2011b) and Chai et al. (2016) and the climate-specific default IPCC indirect N₂O emission factors from the 2019 Refinement to the 2006 IPCC Guidelines are used. For all other livestock the IPCC Tier 1 methodology is used to estimate volatilization. Default parameters are used from the 2006 IPCC Guidelines and 2019 Refinement to the 2006 IPCC Guidelines as described in NIR Annex 3.4.

b. A modified IPCC Tier 1 methodology is used to estimate N₂O emissions from leaching in agricultural soils, as described in NIR Annex 3.4.

Table A6.4–23 Fraction of N Volatilized (FRAC_{GASM}) as Ammonia Resulting from the Application of Biosolid N to Agricultural Soils

IPCC default emission factor, FRAC _{GASM}	0.21 kg NH ₃ -N volatilized kg ⁻¹ N applied
Note: Source: IPCC (2019)	

Table A6.4–24 N₂O Emission Factor for Mid-latitude Cultivation of Organic Soils

	Annual crops	Perennial crops
IPCC default emission factor for mid-latitude cultivation of organic soils	13.0 kg N ₂ O-N ha ⁻¹ yr ⁻¹	9.5 kg N ₂ O-N ha ⁻¹ yr ⁻¹
Note: Source: IPCC (2014)		

Table A6.4–25 Emission Factors (EF) for Biosolid Nitrogen (N) Lost Indirectly as N₂O Due to Leaching of Biosolids Applied to Agricultural Soils

	EF (g N ₂ O kg ⁻¹ N applied)						
	AB	BC	MB	ATL ^b	ON	QC	SK
Leaching^a							
1990	2.0	1.9	2.1	3.5	3.1	3.4	1.5
2005	2.0	2.0	2.1	3.5	3.0	3.4	1.5
2020	2.0	2.0	2.1	3.5	3.0	3.4	1.5
2023	2.0	2.0	2.1	3.5	3.0	3.4	1.5

Notes:

a. A modified IPCC Tier 1 methodology is used to estimate N₂O emissions from leaching in agricultural soils, as described in NIR Annex 3.4.

b. ATL = Atlantic Canada (NB, NL, NS, PE)

Table A6.4–26 Fractions of N Volatilized (FRAC_{GASF}) as Ammonia Resulting from the Application of Inorganic N Fertilizer, from Select Years, 1990–2023, at a Provincial Scale

Year	Implied EF (kg NH ₃ -N volatilized kg ⁻¹ inorganic fertilizer N applied)									
	AB	BC	MB	NB	NL	NS	ON	PE	QC	SK
1990	0.05	0.07	0.05	0.05	0.06	0.06	0.07	0.05	0.07	0.05
2005	0.05	0.08	0.06	0.05	0.06	0.06	0.07	0.05	0.06	0.05
2020	0.05	0.08	0.05	0.05	0.00	0.06	0.06	0.05	0.06	0.05
2023	0.05	0.07	0.05	0.05	0.00	0.06	0.06	0.05	0.07	0.05

Table A6.4–27 Indirect N₂O Emissions from Agricultural Soils

Emission factor due to volatilization and redeposition of Nitrogen ^a	Wet climate: 0.014 kg N ₂ O-N kg ⁻¹ N Dry climate: 0.005 kg N ₂ O-N kg ⁻¹ N
Emission factor due to leaching/runoff ^b	0.0075 kg N ₂ O-N kg ⁻¹ N
Notes: a. Source: IPCC (2019) b. Source: IPCC (2006)	

A6.4.5. Other Sources

Table A6.4–28 CH₄ and N₂O Emissions from Field Burning of Agricultural Residues

CH ₄ emission factor	2.7 g CH ₄ kg ⁻¹ dry matter burnt
N ₂ O emission factor	0.07 g N ₂ O kg ⁻¹ dry matter burnt
Note: Source: IPCC (2006)	

Table A6.4–29 CO₂ Emissions from Liming and Urea Fertilization

Dolomite emission factor	0.13 Mg C / Mg dolomite applied
Limestone emission factor	0.12 Mg C / Mg limestone applied
Urea emission factor	0.20 Mg C / Mg urea
Note: Source: IPCC (2006)	

A6.5. Land Use, Land-Use Change and Forestry

The IPCC Tier 2 and Tier 3 methods and country-specific parameters are used for generating estimates for most of the Land Use, Land-Use Change and Forestry (LULUCF) sector. The Carbon Budget Model of the Canadian Forest Sector or [CBM-CFS3 model](#)³ is used for estimating growth, litter fall, tree mortality and decomposition, as well as the effects of natural disturbances for Forest Land and the emissions due to forest conversion to other land uses. For Cropland, a process model ([CENTURY](#)⁴) is used for estimating CO₂ emissions and removals as influenced by management activities, based on the [National Soil Database](#) of the Canadian Soil Information System.⁵ More details on methods for Forest Land, forest conversion and Cropland are provided in [Annex 3.5](#) of this report, specifically in sections [A3.5.2](#) and [A3.5.4](#).

A country-specific model, the National Forest Carbon Monitoring, Accounting and Reporting System for Harvested Wood Products ([NFCMARS-HWP](#)),⁶ is used to estimate the emissions from the use and disposal of wood products reported under the Harvested Wood Products category. For details on the methods used, see section [A3.5.3](#).

Emissions due to the conversion and management of peatlands for peat extraction, the creation of flooded lands (reservoirs) on areas with no evidence of forest clearing and from the conversion of grasslands to Settlements, are estimated using IPCC Tier 2 methods and country-specific parameters (see sections [A3.5.6.1](#), [A3.5.6.2](#) and [A3.5.7.3](#)). Net CO₂ removals from the growth of urban trees are estimated using an IPCC Tier 2A approach (see section [A3.5.7.1](#)). In addition, emissions due to the occasional burning of grassland are estimated using an IPCC Tier 1 method and default emission factors (see section [A3.5.5.1](#)).

A compilation of emission factors and other parameters used to develop and report the LULUCF estimates is provided in this annex (see [Figure A6.5–1](#) and [Table A6.5–1](#) to [Table A6.5–10](#)).

3 Carbon Budget Model of the Canadian Forest Sector: <https://www.nrcan.gc.ca/climate-change/climate-change-impacts-forests/carbon-accounting/carbon-budget-model/13107>

4 CENTURY Soil Organic Matter Model Environment: <https://www2.nrel.colostate.edu/projects/irc/public/Documents/Software/Century5/Reference/html/Century/cent5-overview.htm>

5 The National Soil Database: <https://sis.agr.gc.ca/cansis/nsdb/index.html>

6 Canada's forest carbon reporting system: <https://natural-resources.canada.ca/climate-change/forest-carbon/canada-s-forest-carbon-reporting-system>

Figure A6.5–1 Disturbance Matrix Parameters for Carbon Modelling (Selected Examples)

Disturbance Matrix Simulating the Carbon Transfers Associated with Clear-Cut Harvesting and Salvage Logging Applicable in all Ecozones Except Those in Alberta and Quebec

	13	14	15	16	17	18	19	24	25	Products
1. Softwood merchantable					0.15					0.85
2. Softwood foliage	1									
3. Softwood others			1							
4. Softwood sub-merchantable			1							
5. Softwood coarse roots			0.5	0.5						
6. Softwood fine roots	0.5	0.5								
7. Hardwood merchantable					0.15					0.85
8. Hardwood foliage	1									
9. Hardwood other			1							
10. Hardwood sub-merchantable			1							
11. Hardwood coarse roots			0.5	0.5						
12. Hardwood fine roots	0.5	0.5								
13. Above-ground very fast soil C	1									
14. Below-ground very fast soil C		1								
15. Above-ground fast soil C			1							
16. Below-ground fast soil C				1						
17. Medium soil C					1					
18. Above-ground slow soil C						1				
19. Below-ground slow soil C							1			
20. Softwood stem snag					0.5					0.5
21. Softwood branch snag			1							
22. Hardwood stem snag					0.5					0.5
23. Hardwood branch snag			1							
24. Black C								1		
25. Peat									1	

Disturbance Matrix Simulating the Carbon Transfers Associated with Firewood Collection from Hardwood applicable in Atlantic Maritime and Mixewood Plains Ecozones

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	Products
1. Softwood merchantable	1																									
2. Softwood foliage		1																								
3. Softwood others			1																							
4. Softwood sub-merch				1																						
5. Softwood coarse roots					1																					
6. Softwood fine roots						1																				
7. Hardwood merch							0.7																			0.3
8. Hardwood foliage								0.7					0.3													
9. Hardwood other									0.7						0.1											0.2
10. Hardwood submerch										0.7													0.3			
11. Hardwood coarse roots											0.7				0.15	0.15										
12. Hardwood fine roots												0.7	0.15	0.15												
13. Above-ground very fast soil C													1													
14. Below-ground very fast soil C														1												
15. Above-ground fast soil C															1											
16. Below-ground fast soil C																1										
17. Medium soil C																	1									
18. Above-ground slow soil C																		1								
19. Below-ground slow soil C																			1							
20. Softwood stem snag																				1						
21. Softwood branch snag																					1					
22. Hardwood stem snag																						0.5				0.5
23. Hardwood branch snag																							0.5			0.5
24. Black C																								1		
25. Peat																									1	

A6.5.2. Harvested Wood Products

Table A6.5–1 Default Parameter Values Used in Harvested Wood Products Analysis

Description	Units	Value
Bark expansion factor, Softwoods	dimensionless	1.11 ^a
Bark expansion factor, Hardwoods	dimensionless	1.15 ^a
Bark expansion factor, Mixedwoods	dimensionless	1.13 ^a
C content of wood	tonnes C od tonne ^{-1, b}	0.5 ^c

Notes:
a. IPCC (2006) (Vol. 4, Table 12.5)
b. Tonnes carbon per oven dry tonne (od tonne) of wood material.
c. IPCC (2006) (Vol. 4, Table 12.4)

Table A6.5–2 Wood Densities of Commodities

Country/Countries	Description	Units	Value
Canada	Species-weighted average density, Roundwood	od tonne m ⁻³	0.386 ^a
Canada	Species-weighted average density, Sawnwood	od tonne m ⁻³	0.481 ^a
Canada	Species-weighted average density, Other industrial roundwood	od tonne m ⁻³	0.583 ^a
Canada	Species-weighted average density, Panels	od tonne m ⁻³	0.643 ^b
Canada	Species-weighted average density, Bioenergy	od tonne m ⁻³	0.523 ^c
U.S.	Coniferous (C) roundwood	od tonne green m ⁻³	0.455 ^d
U.S.	Nonconiferous (NC) roundwood	od tonne green m ⁻³	0.527 ^d
U.S.	C+NC roundwood	od tonne green m ⁻³	0.465 ^d
U.S.	Hardwood (HW) plywood & veneer	tonnes C m ⁻³	0.28 ^e
U.S.	Softwood (SW) lumber	tonnes C m ⁻³	0.22 ^e
U.S.	HW lumber	tonnes C m ⁻³	0.26 ^e
U.S.	Particle board	tonnes C m ⁻³	0.29 ^e
U.S.	Hardboard	tonnes C m ⁻³	0.42 ^e
U.S.	Medium density fibreboard	tonnes C m ⁻³	0.32 ^e
U.S.	Fibreboard, compressed	tonnes C m ⁻³	0.37 ^a
U.S.	Pulp, paper & board	tonnes C ad tonne ⁻¹	0.42 ^e
U.S.	Insulating board	tonnes C m ⁻³	0.45 ^e
All	Sawnwood – C	od tonne m ⁻³	0.45 ^f
All	Sawnwood – NC	od tonne m ⁻³	0.45 ^f
All	Panels, structural	od tonne m ⁻³	0.628 ^f
All	Panels, non-structural	od tonne m ⁻³	0.628 ^f
All	Paper	od tonne ad tonne ⁻¹	0.9 ^f
All	Wood pulp	od tonne ad tonne ⁻¹	0.9 ^f

Notes:

od tonne = Oven dry tonne of wood material

ad tonne = Air dry tonne of product

a. Country-specific values derived from internal analysis of Canada's forest management levels carried out in 2011–2012.

b. Derived from Briggs D (1994)

c. Average of the values for roundwood, sawnwood, other industrial roundwood and panels.

d. FAO (2010)

e. Skog (2008)

f. IPCC (2006) (Vol. 4, Table 12.4)

Table A6.5–3 Half-Life Parameters (Years) of Harvested Wood Products In-use

Country/Countries	Description ^a	Value
Canada	Sawnwood	35 ^b
Canada	Wood panels	25 ^c
Canada	Pulp and paper	2 ^b
Canada	Other industrial roundwood	35 ^b
U.S.	Sawnwood	40 ^b
U.S.	Wood panels	27.04 ^c
U.S.	Pulp and paper	3.25 ^c
U.S.	Other industrial roundwood	40 ^b
Rest of world	Sawnwood	35 ^b
Rest of world	Wood panels	25 ^c
Rest of world	Pulp and paper	2 ^b
Rest of world	Other industrial roundwood	35 ^b

Notes:

a. Firewood and mill residue assumed to be burned for the former, or disposed of for the latter, in the year of harvest.

b. IPCC (2003) (Appendix 3a.1, Table 3a.1.3)

c. Derived from IPCC (2003) (Appendix 3a.1, Table 3a.1.3)

A6.5.3. Cropland

Table A6.5–4 Coefficients of Agricultural Area-based Soil Organic Carbon factors by Land Management Change (LMC)

Zone ^{a, e}	LMC ^{b, c}	k yr ⁻¹	ΔC_{LMCmax} (Mg ha ⁻¹)	Final Year of Effect after LMC ^d	Mean Annual Linear Coefficient over Duration of Effect of LMC (Mg ha ⁻¹ yr ⁻¹)	Mean Annual Linear Coefficient over First 20 Years after LMC (Mg/ha per year)
East Atlantic	IT to NT	0.0216	3.5	52	0.05	0.06
	IT to RT	0.0251	2.4	36	0.04	0.05
	RT to NT	0.0233	1.1	1	0.03	0
	Increase perennial	0.0217	43.4	167	0.25	0.77
East Central	IT to NT	0.025	5	65	0.06	0.1
	IT to RT	0.0261	1.9	25	0.04	0.04
	RT to NT	0.0255	3.2	46	0.05	0.06
	Increase perennial	0.0247	38.2	147	0.25	0.74
Parkland	IT to NT	0.0286	6.5	70	0.08	0.14
	IT to RT	0.0242	2.8	41	0.04	0.05
	RT to NT	0.0263	3.7	51	0.05	0.07
	Increase perennial	0.0233	29.4	142	0.2	0.55
Semi-arid Prairies	IT to NT	0.0261	4.9	63	0.06	0.1
	IT to RT	0.0188	2.3	30	0.03	0.04
	RT to NT	0.0222	2.5	37	0.04	0.05
	Increase perennial	0.0281	26.1	120	0.21	0.56
West	IT to NT	0.0122	4.8	69	0.04	0.05
	IT to RT	0.0116	0.8	0	0	0
	RT to NT	0.0119	3.9	53	0.03	0.04
	Increase perennial	0.0155	34.4	198	0.17	0.46

Notes:

Effective Linear Coefficients of SOC were generated using $F_{LMC(t)} = \Delta C_{LMCmax} \times [1 - \exp(-k \times t)]$.

a. Area-weighted summary: East Atlantic is the Atlantic Maritime reporting zone plus the Boreal Shield reporting zone in Newfoundland and Labrador; East Central is the Mixedwood Plains reporting zone plus the Boreal Shield East reporting zone in Ontario and Quebec; Parkland is the Subhumid Prairies, Boreal Shield West and Boreal Plains reporting zones plus those parts of the Montane Cordillera reporting zone with agricultural activity contiguous to agricultural activity within the rest of the Parkland zone; and West is the Pacific Maritime reporting zone plus the Montane Cordillera reporting zone excepting that portion of the latter that is included in the Parkland zone as described above.

b. For LMCs in the opposite direction to that listed, the ΔC_{LMCmax} will be the negative of the value listed.

c. IT = intensive tillage, RT = reduced tillage, NT = no-till

d. No further C changes once the absolute value of the rate of change is less than 25 kg C ha⁻¹ yr⁻¹.

e. Factors estimated from changes to yield, and as a consequence, carbon inputs to soils cannot be summarized by single regional factors, alternative approaches to presenting simplified factors are under development.

Table A6.5–5 C:N Ratios of Applied Manure from Major Livestock Types and Manure Management Systems on Agricultural Soils

Livestock	Manure Type	C:N Ratio ^a
Dairy Cattle	Liquid	15.0 (± 7.3) ^b
	Solid	13.1 (± 1.4) ^c
Beef Cattle	Solid	14.1 (± 4.7) ^d
Swine	Liquid	10.5 (± 2.4) ^e
Poultry	Solid	5.0 (± 0.4) ^f

Notes:

a. Numbers in the parentheses are standard deviations.

b. Bhandral et al. (2009); Rochette et al. (2008); Pelster et al. (2012); Lynch et al. (2005); Maillard et al. (2015); Maillard et al. (2016); Aoyama et al. (1999)

c. Rochette et al. (2008)

d. Hao et al. (2003); Sommerfeldt et al. (1988); Izaurrealde et al. (2001); Grant et al. (2001); Pare et al. (1999); Larney and Janzen (1996)

e. Pelster et al. (2012); Chantigny et al. (2000); Chantigny et al. (2007); Chantigny et al. (2010); Chantigny et al. (2012)

f. Pelster et al. (2012); Forge et al. (2016); Rees et al. (2014); Larney and Janzen (1996)

Table A6.5–6 Manure-induced C Retention in Soils under annual crop production from Manure Application

Manure Type	Manure-induced C Retention %
Solid Swine and Cattle Manure	25.6 (± 1.4) ^a
Composted Swine Manure	35.5 (± 2.2)
Liquid Swine and Cattle Manure	4.7 (± 3.6)

Note:

a. Standard errors

Data source: Liang et al. (2021)

Table A6.5–7 Soil CO₂ Loss Factor for Mid-latitude Cultivation of Organic Soils

	Annual crops	Perennial crops
IPCC default emission factor for mid-latitude cultivation of organic soils	7.9 Mg CO ₂ -C ha ⁻¹ yr ⁻¹	5.7 Mg CO ₂ -C ha ⁻¹ yr ⁻¹

Note:

Source: IPCC (2014)

Table A6.5–8 Parameters and Emission Factors (EF) for Estimating Emissions from Peat Extraction

Emission Factor/Parameter	Unit ^a	Value
Biomass Clearing		
Forest land biomass cleared	t C ha ⁻¹	19.2 ^b
Other land biomass cleared	t C ha ⁻¹	2.8 ^b
Drainage, Active Extraction, and Unrestored		
CO ₂ from drained, actively extracted, and unrestored areas	t CO ₂ ha ⁻¹ yr ⁻¹	9.42 ^c
CO ₂ -DOC from drained, actively extracted, and unrestored areas	t CO ₂ ha ⁻¹ yr ⁻¹	1.38 ^d
CH ₄ from drained, actively extracted, and unrestored areas	t CH ₄ ha ⁻¹ yr ⁻¹	0.020 ^e
N ₂ O from drained, actively extracted, and unrestored areas	t N ₂ O ha ⁻¹ yr ⁻¹	0.00047 ^f
Peat Stockpiling and Product Production		
Amount of stockpiled peat	t C ha ⁻¹	50 ^g
Exponential decay constant, stockpiled peat		0.05 ^g
Carbon fraction of peat products	t C t air-dry peat ⁻¹	0.26 ^h
Natural Regeneration/Reclamation		
CO ₂ natural regeneration after block-cut	t CO ₂ ha ⁻¹ yr ⁻¹	8.7 ⁱ
CH ₄ natural regeneration after block-cut	t CH ₄ ha ⁻¹ yr ⁻¹	0.012 ^j
CO ₂ natural regeneration after vacuum	t CO ₂ ha ⁻¹ yr ⁻¹	7.6 ^k
CH ₄ natural regeneration after vacuum	t CH ₄ ha ⁻¹ yr ⁻¹	0.043 ^l
CO ₂ -DOC natural regeneration	t CO ₂ ha ⁻¹ yr ⁻¹	0.33 ^m
CO ₂ tree plantation peat efflux	t CO ₂ ha ⁻¹ yr ⁻¹	14.6 ⁿ
Rewetting and Restoration		
CO ₂ MLTT restored ≤12 years	t CO ₂ ha ⁻¹ yr ⁻¹	7.89 ^o
CO ₂ MLTT restored >12 years	t CO ₂ ha ⁻¹ yr ⁻¹	-3.29 ^p
CH ₄ MLTT restored	t CH ₄ ha ⁻¹ yr ⁻¹	0.03 ^q
CO ₂ active rewetting	t CO ₂ ha ⁻¹ yr ⁻¹	-4.50 ^r
CH ₄ active rewetting	t CH ₄ ha ⁻¹ yr ⁻¹	0.19 ^r
CO ₂ -DOC from restored/rewetted Areas	t CO ₂ ha ⁻¹ yr ⁻¹	0.33 ^m
N ₂ O from MLTT restored/rewetted Areas	t N ₂ O ha ⁻¹ yr ⁻¹	N/A ^s
Notes: MLTT = Moss Layer Transfer Technique a. All units where the greenhouse gas (GHG) is specified use units of the relevant GHG (CO ₂ , CH ₄ or N ₂ O) instead of carbon (C) or nitrogen (N). b. Hayne and Verbicki (2011) c. Moore et al. 2002, as cited in Cleary 2003; Glatzel et al. 2003; Waddington et al. 2010; Strack & Zuback 2013; Strack et al. 2014; Nugent et al. 2019; Rankin et al. 2018; Clark et al. 2023; Turmel-Courchesne et al. 2023; Hunter et al. 2024 d. Waddington et al. (2008), Strack and Zuback (2013) e. Moore et al. (2002) (unpublished data) as cited in Cleary (2003); Waddington and Day (2007); Strack and Zuback (2013); Strack et al. (2014); Rankin et al. (2018); Nugent et al. (2019); Turmel-Courchesne et al. (2023); Clark et al. (2023); Hunter et al. (2024) f. IPCC, 2014 (Table 2.5, Default value for Boreal & Temperate climate zone) g. Cleary (2003); Hayne et al. (2014); Sharma et al. (2024) h. Hayne et al. (2014) i. Waddington and Price (2000); Waddington et al. (2002); Waddington and Warner (2001); McNeil and Waddington (2003); Cagampan and Waddington (2008) j. Waddington and Price (2000) k. Moore et al. (2002) as cited in Cleary (2003); Petrone et al. (2001); Petrone et al. (2003); Waddington et al. (2010); Strack and Zuback (2013); Strack et al. (2014) l. Moore et al. (2002) as cited in Cleary (2003); Waddington and Day (2007); Strack and Zuback (2013); Strack et al., (2014) m. Waddington et al. (2008); Strack and Zuback (2013); Nugent et al. (2018); D'Acuna et al., (2019) n. Garcia Bravo et al. (2018) o. Moore et al. (2002) (unpublished data) as cited in Cleary (2003); Petrone et al. (2001); Petrone et al. (2003); Waddington et al. (2010); Strack & Zuback (2013); Strack et al. (2014); Nugent et al. (2019); Lazzano et al. (2018) p. Nugent et al. (2018) q. Moore et al. (2002) (unpublished data) as cited in Cleary (2003); Waddington and Day (2007); Strack and Zuback (2013); Strack et al. (2014); Nugent et al. (2018); Nugent et al. (2019); Lazzano et al. (2018) r. Turmel-Courchesne et al. (2023); Lee et al. (2017); D'Acunha et al. (2019) s. IPCC 2014 Wetlands Supplement, Default assumption of no N ₂ O emissions from rewetted/restored areas		

A6.5.5. Settlements

Table A6.5–9 Carbon Storage and Sequestration in Urban Trees		
Reconciliation Unit (RU)	Carbon Storage (t C ha ⁻¹)	Net Carbon Sequestration (t C ha ⁻¹ yr ⁻¹)
1 NL – Boreal Shield East	40	3.0
5 NS – Atlantic Maritime	62	3.4
6 PE – Atlantic Maritime	62	3.4
7 NB – Atlantic Maritime	62	3.4
11 QC – Atlantic Maritime	62	3.4
12 QC – Mixedwood Plains	58	2.4
15 QC – Boreal Shield East	40	3.0
16 ON – Boreal Shield West	40	3.0
17 ON – Mixedwood Plains	58	2.4
19 ON – Boreal Shield East	40	3.0
24 MB – Subhumid Prairies	55	2.9
28 SK – Boreal Plains	40	3.0
30 SK – Semiarid Prairies	55	2.9
34 AB – Boreal Plains	40	3.0
35 AB – Subhumid Prairies	55	2.9
37 AB – Semiarid Prairies	55	2.9
41 BC – Pacific Maritime	97	6.9
42 BC – Montane Cordillera	23	1.4
Note: Source: Steenberg et al., 2021		

A6.5.6. Forest Conversion

Table A6.5–10 Soil Organic Carbon in Forest and Agricultural Land (0- to 30-cm soil depth)			
Soil Texture	Soil Organic Carbon (Mg C ha ⁻¹)		Difference (%)
	Forested Land ^a	Cropland ^a	
Eastern Canada			
Coarse	85 (26)	68 (42)	-20
Medium	99 (38)	77 (35)	-22
Fine	99 (58)	78 (36)	-21
Western Canada			
Coarse	73 (39)	74 (38)	0
Medium	66 (30)	73 (30)	4
Fine	74 (38)	77 (25)	1
Note: a. Standard deviation in parentheses.			

Table A6.5–11 Parameters and Emission Factors (EF) for Estimating Emissions from Northern Land Use Change (NLUC)

Emission Factor/Parameter		
Above-ground Biomass (AGB)		
	Unit	Value
Landcover of Canada Vegetation Class^a		
Temperate or sub-polar needleleaf forest	t C ha ⁻¹	24 ^{b,c}
Sub-polar taiga needleleaf forest	t C ha ⁻¹	17 ^{b,c}
Temperate or sub-polar broadleaf deciduous forest	t C ha ⁻¹	24 ^{b,c}
Mixed forest	t C ha ⁻¹	24 ^{b,c}
Temperate or sub-polar shrubland	t C ha ⁻¹	10 ^b
Temperate or sub-polar grassland	t C ha ⁻¹	4.8 ^b
Sub-polar or polar shrubland-lichen-moss	t C ha ⁻¹	8 ^b
Sub-polar or polar grassland-lichen-moss	t C ha ⁻¹	4.8 ^b
Sub-polar or polar barren-lichen-moss	t C ha ⁻¹	1.5 ^{b,d}
Wetland	t C ha ⁻¹	3.8 ^b
Below-ground Biomass (BGB)		
Ecozone		
Boreal	BGB:AGB ratio	0.21 ^e
Arctic	BGB:AGB ratio	0.69 ^e

Notes:
 Values are derived from a Delphi expert consultation (Lapointe, 2021) and primary literature (Wulder et al., 2020; Bona et al., 2020), where the average values from matches expert opinion.
 a. Latifovic (2019)
 b. Lapointe (2021)
 c. Wulder et al. (2020)
 d. Bona et al. (2018)
 e. Poorter et al. (2012)

A6.6. Biomass Combustion

A6.6.1. CO₂

Emissions of CO₂ from the combustion of biomass originated from forests or other woodlands (whether for energy use, from prescribed burning or from wildfires) are not included in National Inventory totals. Emissions from prescribed burning and from the combustion of this biomass for energy use are estimated and reported in the Land Use, Land-use Change and Forestry (LULUCF) sector, in common reporting tables (CRT) 4(V) and 4.G respectively. Forest wildfires are considered uncontrollable natural disturbances in the modelling and reporting approach used in the LULUCF sector by which these emissions and subsequent removals are estimated and tracked separately from emissions/removals resulting from commercially managed forest stands, more details on his approach can be found in section [A3.5.2.4](#).

The CO₂ emissions related to the combustion of biomass for energy use are reported as memo items in the CRT of the Energy sector, as required by the modalities, procedures and guidelines for the transparency framework for action and support referred to in Article 13 of the Paris Agreement (MPGs).

Emissions from industrial combustion of biomass are dependent primarily on the characteristics of the fuel being combusted. The CO₂ emission factor (EF) ([Table A6.6–1](#)) for industrial wood waste has been developed from facility source sampling data collected by the U.S. EPA in units of lb/MMBTU (one million British thermal units; U.S. EPA, 2003). The U.S. EPA data were converted to kg/tonne at 0% moisture content (m.c.) using a higher heating value (HHV) of 20.44 MJ/kg, which was developed from an internal review of available moisture content and heating value data. The EF for spent pulping liquor is calculated from data collected by the National Council for Air and Stream Improvement (NCASI), based on carbon content assuming a 1% correction for unoxidized carbon (NCASI, 2010). The NCASI EFs were reported in units of kg/GJ HHV, which was converted to kg/tonne at 0% m.c. using a HHV of 13.7 MJ/kg (Tran, 2014). The EF for biogenic waste fuels used in the cement industry is calculated from fuel use data reported by the cement industry, using CO₂ accounting and reporting standards developed by the World Business Council for Sustainable Development (WBSCD, 2005) for years 2001-2016. From 2017 onwards, fuel use data and emissions were reported by cement facilities through the Greenhouse Gas Reporting Program (GHGRP) (ECCC, 2024a). An weighted average EF was developed based on reported data.

CO₂ EFs for residential combustion (Table A6.6–1) are based on an ECCC study of emissions from wood burning appliances (ECCC, 2024b). The study collected emissions data from residential wood burning appliances that are typical for Canada and represent improvements to pollution control over the time series. The EFs are reported at a 25% m.c., except for the EPA certified pellet stove, which is reported at a 10% m.c.

CO₂ emissions occur during forest wildfires and from controlled burning during forest conversion activities. The carbon emitted as CO₂ (CO₂-C) during forest fires is considered in the forest carbon balance, whereas the CO₂-C emitted during controlled burns is reported under the new land-use categories. There is no unique CO₂ EF applicable to all fires, as the proportion of CO₂-C emitted for each pool can be specific to the pool, the type of forest and disturbance, and the ecological zone (see Annex 3.5).

EFs for gaseous biomass (Table A6.6–2), which includes landfill gas, sludge gas from wastewater treatment and digester gas from centralized anaerobic digesters, are adapted from the IPCC (2006).

Table A6.6–1 Emission Factors for Biomass					
Source ^a	Description	Emission Factor			
		CO ₂	CH ₄	N ₂ O	Units
Wood Fuel / Wood Waste	Industrial Combustion	1 715 ^b	0.1 ^c	0.07 ^c	g/kg fuel
Forest Wildfires	Open Combustion	NA	NA ^d	NA ^e	g/kg fuel
Controlled Burning	Open Combustion	NA	NA ^d	NA ^e	g/kg fuel
Spent Pulping Liquor	Industrial Combustion	1 250 ^f	0.03 ^g	0.005 ^g	g/kg fuel
Stoves and Fireplaces	Residential Combustion				
Open Fireplace		1 162 ^h	5.04 ^h	0.057 ^h	g/kg fuel
Conventional Woodstove		1 096 ^h	7.85 ^h	0.065 ^h	g/kg fuel
Fireplace Insert		1 212 ^h	5.70 ^h	0.062 ^h	g/kg fuel
Pre-2020 EPA Certified Woodstove		1 206 ^h	4.11 ^h	0.067 ^h	g/kg fuel
2020 EPA Certified Non-Catalytic Woodstove		1 195 ^h	4.28 ^h	0.058 ^h	g/kg fuel
2020 EPA Certified Catalytic Woodstove		1 203 ^h	7.70 ^h	0.065 ^h	g/kg fuel
EPA Certified Pellet Stove		1 449 ^h	1.74 ^h	0.069 ^h	g/kg fuel
Cement Industry Waste Fuel (biogenic)	Industrial Combustion	86 ⁱ	0.03 ^j	0.004 ^j	kg/GJ

Notes:

NA = Not applicable

a. CO₂ emissions from biomass combusted for energy or agricultural purposes are not included in inventory totals, whereas CH₄ and N₂O emissions from these sources are inventoried under the Energy Sector. All greenhouse gas (GHG) emissions, including CO₂ emissions from biomass burned in managed forests (wildfires and controlled burning), are reported under Land-Use, Land-use Change and Forestry (LULUCF) and excluded from national inventory totals.

b. Adapted from U.S. EPA (2003)

c. Adapted from U.S. EPA (2003) and NCASI TB998 (2012)

d. Emission ratio for CH₄ is 1/90th CO₂. See NIR Annex 3.4.

e. Emission ratio for N₂O is 0.017% CO₂. See NIR Annex 3.4.

f. Adapted from NCASI (2011)

g. Adapted from NCASI (2012)

h. ECCC (2024b)

i. Adapted from WBSCD (2005) and ECCC (2024a)

j. Adapted from IPCC (2006)

A6.6.2. CH₄

Emissions of CH₄ from residential combustion of firewood are technology-dependent. The CH₄ EFs are based on an ECCC study of emissions from wood burning appliances (ECCC, 2024b). The EFs are reported at a 25% m.c., except for the EPA certified pellet stove, which is reported at a 10% m.c.

Emissions from industrial combustion of biomass are dependent primarily on the characteristics of the fuel being combusted. The EF (Table A6.6–1) for CH₄ from industrial wood waste has been developed from facility source sampling data collected by the U.S. EPA in units of lb/MMBTU (U.S. EPA, 2003) and collected by the NCASI in units of kg/MMBTU and converted to kg/tonne at 0% m.c. as discussed in section A6.6.1. The EF for CH₄ from spent pulping liquor has been developed using source sampling data from NCASI in units of kg/MMBTU, converted to kg/tonne at 0% m.c. using a HHV of 13.7 MJ/kg as discussed in section A6.6.1. The CH₄ EF for bio-based alternative fuels (Table A6.6–1) was adapted from the 2006 IPCC Guidelines (IPCC, 2006).

EFs from gaseous biomass (Table A6.6–2), which includes landfill gas, sludge gas from wastewater treatment and digester gas from centralized anaerobic digesters, are adapted from the IPCC (2006).

Emissions of carbon as CH₄ (CH₄-C) from wildfires and controlled burning are always equal to 1/90th of CO₂-C emissions.

A6.6.3. N₂O

Emissions of N₂O from residential combustion of firewood are technology-dependent. The N₂O EFs are based on an ECCC study of emissions from wood burning appliances (ECCC, 2024b). The EFs are reported at a 25% m.c., except for the EPA certified pellet stove, which is reported at a 10% m.c.

Emissions from industrial combustion of biomass are dependent primarily on the characteristics of the fuel being combusted. EFs (Table A6.6–1) for industrial wood waste has been developed from facility source sampling data collected by the U.S. EPA in units of lb/MMBTU (U.S. EPA, 2003) and collected by the National Council for Air and Stream Improvements (NCASI) in units of kg/MMBTU and converted to kg/tonne at 0% m.c. as discussed in section A6.6.1. The EF for N₂O from spent pulping liquor has been developed using source sampling data from NCASI in units of kg/MMBTU, converted to kg/tonne at 0% m.c. using a HHV of 13.7 MJ/kg as discussed in section A6.6.1. The N₂O EF for bio-based alternative fuels (Table A6.6–1) was adapted from the 2006 IPCC Guidelines (IPCC, 2006).

EFs for gaseous biomass (Table A6.6–2), which includes landfill gas, sludge gas from wastewater treatment and digester gas from centralized anaerobic digesters, are adapted from the IPCC (2006).

N₂O emissions from wildfires and controlled burning are equal to 0.017% vol/vol of CO₂ emissions. Since both gases have the same molecular weight, the same ratio can be applied on a mass basis (see section A3.5.2).

Table A6.6–2 **Emission Factors for Gaseous Biomass Combustion**

Source	Description	Emission Factor			Emission Factor Units
		CO ₂	CH ₄	N ₂ O	
Landfill Gas	Industrial Combustion (for Energy)	2 752 ^a	0,05 ^a	0,005 ^a	kg/t CH ₄ Utilized for Energy
Sludge Gas	Industrial Combustion (for Energy)	2 752 ^a	0,05 ^a	0,005 ^a	kg/t CH ₄ Utilized for Energy
Digester Gas	Industrial Combustion (for Energy)	2 752 ^a	0,05 ^a	0,005 ^a	kg/t CH ₄ Utilized for Energy
Landfill Gas	Flaring	NE	3.0 ^b	NE	kg/t CH ₄ Flared

Notes:

NE = Not estimated

a. Adapted from IPCC (2006) (Vol. 2, Table 2.2)

b. U.S. EPA (1998)

A6.7. Waste

A6.7.1. Municipal Wastewater Treatment and Discharge

A6.7.1.1. CH₄

Emissions from municipal wastewater treatment and discharge are dependent on the organic loading of the effluent stream (which is a function of population), and the type of wastewater treatment provided. Emission factors (EF) are the product of the methane correction factor (MCF), which is the technology-specific estimate of the fraction of biological oxygen demand (BOD) that will ultimately degrade anaerobically, and the maximum methane producing capacity (B_0), which is expressed in terms of kg CH₄/kg BOD removed. The IPCC default value of 0.6 kg CH₄/kg BOD for B_0 was not used. The AECOM (2011) study commissioned by Environment Canada confirmed that its derivation from the 0.25 kg CH₄/kg COD was erroneous, where COD is the chemical oxygen demand. A Canada-specific value of 0.36 CH₄/kg BOD for B_0 was used (AECOM, 2011).

The MCF and EF values for CH₄ emissions from wastewater treatment and discharge, by treatment technology are shown in [Table A6.7–1](#).

Table A6.7–1 Methane Correction Factors (MCF) and Emission Factors (EF) for CH₄ from Wastewater Treatment and Discharge

Treatment	MCF	EF
No Treatment	0.1 ^a	0.036
Primary	0.018 ^b	0.0036
Aerobic Lagoon	0 ^a	0
Anaerobic Lagoon	0.8 ^a	0.288
Facultative Lagoon	0.2 ^a	0.072
Other / Unspecified Lagoon	0.2 ^c	0.072
Secondary Anaerobic	0.8 ^a	0.288
Secondary Activated Sludge	0.01 ^b	0.0036
Trickling Filter	0.01 ^b	0.0036
Trickling Filter – High Rate	0.01 ^b	0.0036
Rotating Biological Contactor	0.01 ^b	0.0036
Sequencing Batch Reactor	0.05 ^d	0.018
Secondary Biofiltration	0.01 ^b	0.0036
Secondary with Biological Nutrient Removal	0.01 ^b	0.0036
Septic	0.5 ^a	0.18
Septic with Marine Outfall	0.5 ^a	0.18
Wetland	0.17 ^e	0.0612
Other / Unknown	0.2 ^f	0.072

Notes:

EF calculated as MCF x B_0 , where B_0 is Canada-specific value of 0.36

a. IPCC (2006) (Vol. 5, Chapter 6, Table 6.3)

b. IPCC (2019)

c. Assuming Unspecified lagoon types were assumed to be facultative lagoons.

d. Taseli (2018). Point source pollution and climate change impact from Sequential Batch Reactor wastewater treatment plant.

e. IPCC Supplement to 2006 Guidelines for Wetlands (2014). Chapter 6. Mean value of MCF's for three wetland types provided in document.

f. Assuming facilities of unknown or other treatment type are either facultative lagoon or untreated discharge to sea. The median value of the MCF and EF of these technologies used.

A6.7.1.2. N₂O

N₂O emissions from wastewater are a function of the nitrogen entering the wastewater stream, which is, in turn, a function of protein consumption per capita, population, nitrogen content in protein, and adjustment factors for input of non-consumed nitrogen (e.g., from washing) and industrial inputs. The emission factor used is the IPCC 2006 Guideline default value of 0.005 kg N₂O-N/kg N (IPCC, 2006). The emission factor for N₂O from wastewater treatment and discharge is shown in Table A6.7–2.

Table A6.7–2 **Emission Factors for N₂O from Wastewater Treatment and Discharge**

Treatment Category	EF (kg N ₂ O-N/kg N)	Units
No Treatment	0 ^a	kg N ₂ O-N/kg N ^d
Primary	0 ^b	
Aerobic Lagoon	0 ^b	
Anaerobic Lagoon	0 ^a	
Facultative Lagoon	0 ^b	
Other / Unspecified Lagoon	0 ^c	
Secondary Anaerobic	0 ^a	
Secondary Activated Sludge	0.016 ^a	
Trickling Filter	0.016 ^a	
Trickling Filter – High Rate	0.016 ^a	
Rotating Biological Contactor	0.016 ^a	
Sequencing Batch Reactor	0.016 ^a	
Secondary Biofiltration	0.016 ^b	
Secondary with Biological Nutrient Removal	0.016 ^a	
Septic	0.0045 ^a	
Septic with Marine Outfall	0 ^a	
Wetland	0 ^b	
Other / Unknown	0.016 ^c	
Notes: a. IPCC (2019) b. ECCC Estimate c. Modelled as facultative lagoon d. To convert N ₂ O-N to N ₂ O, multiply by 44/28		

A6.7.2. Waste Incineration

The emission factors for CO₂, CH₄, and N₂O for waste incineration are shown in [Table A6.7–3](#).

Category	Emission Factors – kg / tonne waste		
	CO ₂	CH ₄	N ₂ O
Municipal Solid Waste Incineration – Continuous – Fluidized Bed	3 666.67 ^a	0.0002	0.00005
Municipal Solid Waste Incineration – Continuous – Stoker		0	0.00005
Municipal Solid Waste Incineration – Semi-Continuous – Fluidize Bed		0.006	0.00005
Municipal Solid Waste Incineration – Semi-Continuous – Stoker		0.188	0.00005
Municipal Solid Waste Incineration – Batch – Fluidized Bed		0.06	0.00006
Municipal Solid Waste Incineration – Batch – Stoker		0.237	0.00006
Sewage Sludge Incineration	1 650.00	9.70	0.99
Hazardous Waste Incineration	1 650.00	0.17	3.16
Clinical Waste Incineration – Continuous	572.00	0.0002	0.05
Clinical Waste Incineration – Batch		0.06	0.06
Industrial Sludge – Continuous	1 650.00	0.0002	0.45
Industrial Sludge – Batch		0.06	0.45
Fossil Liquid Waste – Continuous	2 933.33	0.0002	0.1
Fossil Liquid Waste – Batch		0.06	0.1
Fossil Liquid Waste – Semi-Continuous		0.0301	0.1
Industrial – Continuous – Stoker	1 650.00	0.0002	0.05
Industrial – Continuous – Fluidized Bed		0	0.05
Industrial – Semi-Continuous – Fluidized Bed		0.006	0.05
Industrial – Semi-Continuous – Stoker		0.188	0.05
Industrial – Batch – Fluidized Bed		0.06	0.06
Industrial – Batch – Stoker		0.237	0.06

Notes:
Source: IPCC (2006)
a. For Municipal Solid Waste, the emission factor for CO₂ is in units of kg CO₂ / tonne fossil carbon.

A6.7.2.1. Sewage Sludge Incinerators

Emissions from sewage sludge incinerators are estimated from an emission factor obtained from the IPCC 2006 Guidelines (IPCC, 2006).

A6.7.2.2. Municipal Solid Waste Incinerators

The emission estimates from municipal solid waste incineration are calculated based on batch or continuous operations, and based on stoker or fluidized bed combustion technology. The emission factors used are from the IPCC 2006 Guidelines (IPCC, 2006). For CO₂ emissions, only the non-biogenic (fossil) portion of the waste is included when calculating emissions.

A6.7.2.3. Hazardous Waste Incinerators

The CO₂ emission factors for hazardous waste incineration are taken from the IPCC 2006 Guidelines (IPCC, 2006). The CO₂ emission factor is based on a carbon content of 50% and fossil carbon content of 90% of the carbon content. CH₄ and N₂O emission factors were developed for Canada using emissions from a facility.

A6.7.2.4. Clinical Waste Incinerators

The emission factors for clinical waste incineration are taken from the IPCC 2006 Guidelines (IPCC, 2006). The CO₂ emission factor is based on a carbon content of 45%.

A6.7.3. Biological Treatment of Solid Waste

The emission factors for CO₂, CH₄, and N₂O for the biological treatment of solid waste are shown in [Table A6.7–4](#).

Table A6.7–4 Emission Factors for the Biological Treatment of Solid Waste					
	Waste of Facility Type	Emission Factors			Units
		CO ₂	CH ₄	N ₂ O	
Anaerobic Digestion	Off-farm facilities ^a	NA	2.10	NA	% of methane produced in biogas
Composting	Yard Waste ^b	NA	1.72	0.25	g/kg Wet Waste
	Biosolids or Manure ^b	NA	3.54	0.18	g/kg Wet Waste
	Mixture of Wastes ^b	NA	1.09	0.11	g/kg Wet Waste
	Municipal Solid Waste ^b	NA	1.51	0.18	g/kg Wet Waste
Notes: NA = Not applicable a. ECCC(2020a) b. ECCC(2020b)					

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Annex 6, Emission Factors

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